MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

ENCODER, PRIORITY, DIGITAL, KEY CODE MHL

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic priority encoder module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.809 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be MHL.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/76-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

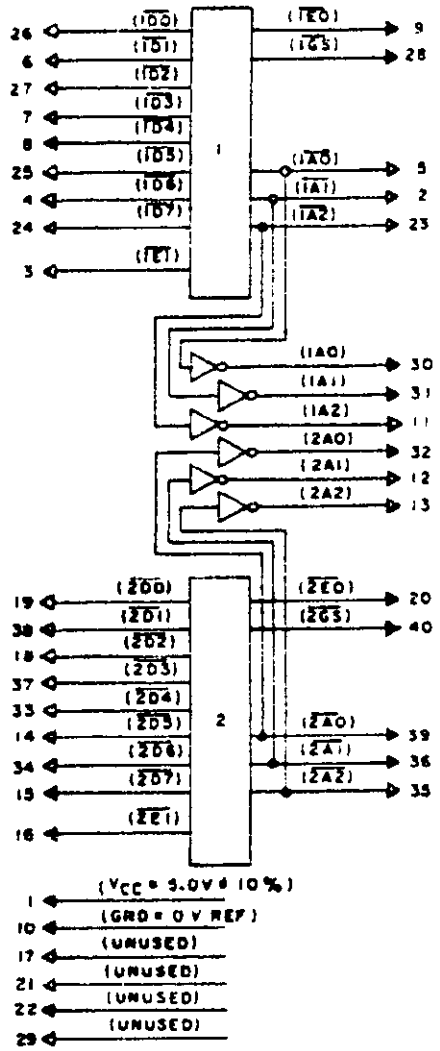
3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage ($V_{CC}$) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

3.10 Functional tests. All modules shall be subjected to functional tests which conform to the requirements steted in 3.10.1 through 3.10.3.

3.10.1 Test patterns. A set of unique input patterns shall be generated capable of detecting all nonredundant failures in the module logic network. The test patterns shall be generated by assuming "stuck at H state" and "stuck at L state" failures in each of the basic logic functions and applying input patterns to detect these failures. The unique test patterns necessary to satisfy these requirements shall be submitted with the qualification samples. If the tests of tables I and VII satisfy or exceed these unique test patterns, no additional tests shall be added; otherwise, the needed tests shall be added to table VII. "Stuck at H state" indicates a failure resulting in a steady high-level output. "Stuck at L state" indicates a failure resulting in a steady low-level output. Prior to any change in module design which would necessitate a different set of test patterns to detect a "stuck at H state" or a "stuck at L state", the vendor shall be required to submit to the acquiring activity and SEMP Quality Assurance Activity the proposed new set of test patterns for approval.



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NOTES:
1. $\overline{1D0}$-$\overline{1D7}$, $\overline{2D0}$-$\overline{2D7}$ = Priority (active low) inputs.
2. $\overline{1E1}$, $\overline{2E1}$ = Enable (active low) inputs.
3. $\overline{1E0}$, $\overline{2E0}$ = Enable (active low) outputs.
4. $\overline{1G5}$, $\overline{2G5}$ = Group select (active low) inputs.
5. $\overline{1A0}$-$\overline{1A2}$, $\overline{2A0}$-$\overline{2A2}$ = Address (active low) outputs.
6. 1A0-1A2, 2A0-2A2 = Address (active high) outputs.

FIGURE 1. MHL module function diagram, priority encoder.



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TABLE I. Module electrical requirements.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | Acceptance requirements 25°C ±5°C ambient temperature | | | | Operating temperature 0°C to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| A1 | Isolation, frame | III | S | 10 | -- | 1 | -- | 10 | -- | 1 | -- | MΩ |
| A2-A5 | Isolation, unused pins | III | X | 10 | -- | 1 | -- | 10 | -- | 1 | -- | MΩ |
| B1-B16 | $V_{OUT}$ (low loaded) | IV | S | 0.01 | 0.80 | 0.01 | 0.80 | 0.01 | 0.80 | 0.01 | 0.80 | V |
| C1-C18 | Input breakdown current | V | S | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | mA |
| D1-D10 | Output short circuit current | VI | X | -30 | -100 | -30 | -100 | -30 | -100 | -30 | -100 | mA |
| D11-D16 | Output short circuit current | VI | X | -20 | -55 | -20 | -55 | -20 | -55 | -20 | -55 | mA |
| E1-E35 | Functional | VII | X | Test vectors must produce output vectors as specified in table VII. | | | | | | | | |
| F1-F18 | Input clamp diode voltage | VIII | X | -0.4 | -1.5 | -0.4 | -1.5 | -0.4 | -1.5 | -0.4 | -1.5 | V |
| G1-G16 | Output voltage, low level | IX | X | 0.0 | 0.4 | 0.0 | 0.4 | 0.0 | 0.4 | 0.0 | 0.4 | V |
| H1-H32 | Output voltage, high level | X | X | 2.4 | 4.5 | 2.4 | 4.5 | 2.4 | 4.5 | 2.4 | 4.5 | V |
| J1,J2 | Input current, high level | XI | X | 0 | 40 | 0 | 40 | 0 | 40 | 0 | 40 | µA |
| J3-J18 | Input current, high level | XI | X | 0 | 80 | 0 | 80 | 0 | 80 | 0 | 80 | µA |
| K1,K2 | Input current, low level | XII | X | -0.10 | -1.73 | -0.10 | -1.73 | -0.10 | -1.73 | -0.10 | -1.73 | mA |
| K3-K18 | Input current, low level | XII | X | -0.20 | -3.46 | -0.20 | -3.46 | -0.20 | -3.46 | -0.20 | -3.46 | mA |
| L1 | Power supply current, low-level outputs | XIII | X | 77 | 187 | 77 | 187 | 77 | 187 | 77 | 187 | mA |
| L2 | Power supply current, high-level outputs | XIII | X | 77 | 187 | 77 | 187 | 77 | 187 | 77 | 187 | mA |
| M1-M16 | Transition time ($t_{TLH}$), $C_L$ = 50 pF | XIV | S | 4 | 30 | 4 | 30 | 4 | 35 | 4 | 35 | ns |
| M17-M32 | Transition time ($t_{TLH}$), $C_L$ = 150 pF | XIV | Q | 4 | 50 | 4 | 50 | 4 | 55 | 4 | 55 | ns |
| N1-N16 | Transition time ($t_{THL}$), $C_L$ = 50 pF | XIV | S | 4 | 13 | 4 | 13 | 4 | 18 | 4 | 18 | ns |

See footnotes at end of table.





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TABLE I. Module electrical requirements - Continued.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | Acceptance requirements 25°C ±5°C ambient temperature | | | | Operating temperature 0°C to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| N17-N32 | Transition time ($t_{THL}$), $C_L$ = 150 pF | XIV | Q | 4 | 20 | 4 | 20 | 4 | 25 | 4 | 25 | ns |
| P1-P8 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | XIV | S | 2 | 12 | 2 | 12 | 2 | 14 | 2 | 14 | ns |
| P9,P10 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | XIV | S | 5 | 18 | 5 | 18 | 5 | 21 | 5 | 21 | ns |
| P11-P20 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | XIV | S | 8 | 44 | 8 | 44 | 8 | 46 | 8 | 46 | ns |
| P21,P22 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | XIV | S | 4 | 19 | 4 | 19 | 4 | 22 | 4 | 22 | ns |
| P23-P56 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | XIV | S | 8 | 33 | 8 | 33 | 8 | 39 | 8 | 39 | ns |
| P57-P62 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | XIV | S | 7 | 21 | 7 | 21 | 7 | 28 | 7 | 28 | ns |
| P63-P74 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | XIV | S | 13 | 54 | 13 | 54 | 13 | 58 | 13 | 58 | ns |
| P75-P80 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | XIV | S | 10 | 51 | 10 | 51 | 10 | 58 | 10 | 58 | ns |
| P81-P88 | Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | XIV | Q | 4 | 17 | 4 | 17 | 4 | 19 | 4 | 19 | ns |
| P89,P90 | Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | XIV | Q | 5 | 24 | 5 | 24 | 5 | 27 | 5 | 27 | ns |
| P91-P100 | Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | XIV | Q | 8 | 50 | 8 | 50 | 8 | 52 | 8 | 52 | ns |
| P101,P102 | Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | XIV | Q | 4 | 25 | 4 | 25 | 4 | 28 | 4 | 28 | ns |
| P103-P136 | Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | XIV | Q | 10 | 39 | 10 | 39 | 10 | 45 | 10 | 45 | ns |

See footnotes at end of table.





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TABLE I. Module electrical requirements - Continued.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | Acceptance requirements 25°C ±5°C ambient temperature | | | | Operating temperature 0°C to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| P137-P142 | Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | XIV | Q | 7 | 27 | 7 | 27 | 7 | 34 | 7 | 34 | ns |
| P143-P154 | Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | XIV | Q | 13 | 66 | 13 | 66 | 13 | 70 | 13 | 70 | ns |
| P155-P160 | Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | XIV | Q | 10 | 63 | 10 | 63 | 10 | 70 | 10 | 70 | ns |
| Q1-Q10 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | XIV | S | 4 | 28 | 4 | 28 | 4 | 30 | 4 | 30 | ns |
| Q11,Q12 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | XIV | S | 8 | 39 | 8 | 39 | 8 | 41 | 8 | 41 | ns |
| Q13-Q22 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | XIV | S | 8 | 26 | 8 | 26 | 8 | 28 | 8 | 28 | ns |
| Q23,Q24 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | XIV | S | 7 | 27 | 7 | 27 | 7 | 31 | 7 | 31 | ns |
| Q25-Q58 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | XIV | S | 8 | 30 | 8 | 30 | 8 | 32 | 8 | 32 | ns |
| Q59-Q64 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | XIV | S | 7 | 27 | 7 | 27 | 7 | 32 | 7 | 32 | ns |
| Q65-Q76 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | XIV | S | 13 | 54 | 13 | 54 | 13 | 62 | 13 | 62 | ns |
| Q77-Q82 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | XIV | S | 10 | 42 | 10 | 42 | 10 | 51 | 10 | 51 | ns |
| Q83-Q92 | Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | XIV | Q | 4 | 33 | 4 | 33 | 4 | 35 | 4 | 35 | ns |

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

| Test condition | Parameter | Ref test table | Test type 1/ - 3/ | Acceptance requirements 25°C ±5°C ambient temperature | | | | Operating temperature 0°C to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| Q93,Q94 | Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | XIV | Q | 10 | 44 | 10 | 44 | 10 | 46 | 10 | 46 | ns |
| Q95-Q104 | Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | XIV | Q | 8 | 31 | 8 | 31 | 8 | 33 | 8 | 33 | ns |
| Q105,Q106 | Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | XIV | Q | 7 | 32 | 7 | 32 | 7 | 36 | 7 | 36 | ns |
| Q107-Q140 | Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | XIV | Q | 10 | 35 | 10 | 35 | 10 | 37 | 10 | 37 | ns |
| Q141-Q146 | Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | XIV | Q | 7 | 32 | 7 | 32 | 7 | 37 | 7 | 37 | ns |
| Q147-Q158 | Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | XIV | Q | 13 | 68 | 13 | 68 | 13 | 76 | 13 | 76 | ns |
| Q159-Q164 | Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | XIV | Q | 10 | 56 | 10 | 56 | 10 | 65 | 10 | 65 | ns |
| R1,R9 | Output crosstalk, low level | XV | Q | 0 | 800 | 0 | 800 | 0 | 800 | 0 | 800 | mV |
| R2-R8, R10-R16 | Output crosstalk, low level | XV | Q | 0 | 650 | 0 | 650 | 0 | 650 | 0 | 650 | mV |
| R17-R32 | Output crosstalk, duration | XV | Q | 0 | 400 | 0 | 400 | 0 | 400 | 0 | 400 | mV |
| S1-S16 | Output crosstalk, high level | XV | Q | 0 | 400 | 0 | 400 | 0 | 400 | 0 | 400 | mV |
| T | Power supply transient amplitude | XVI | Q | 0 | 400 | 0 | 400 | 0 | 400 | 0 | 400 | mV |

See footnotes at end of table.



MIL-M-28787/46(EC)

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TABLE I. Module electrical requirements - Continued.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | Acceptance requirements 25°C ±5°C ambient temperature | | | | Operating temperature 0°C to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| U | Power supply filter capacitance | XVII | S | 1.6 | 3.0 | 1.6 | 3.0 | 1.6 | 3.0 | 1.6 | 3.0 | µF |
| V1-V18 | Input capacitance | XVIII | Q | 5 | 15 | 5 | 15 | 5 | 15 | 5 | 15 | pF |

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.





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3.10.2 Input requirements. The following input voltage levels shall be used for test patterns:

Low level - Input voltage shall be 0.8 V ±1%.
High level - Input voltage shall be 2.0 V ±1%.

3.10.3 Output requirements. The output requirements for the test patterns shall be as follows:

Low level - Output voltage shall be 0 V minimum, 0.4 V maximum.
High level - Output voltage shall be 2.4 V minimum, $V_{CC}$ maximum.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through V shall be performed in any order with the following exceptions. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through V in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through U shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition U, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-R-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through U shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through V shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.



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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

5.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

*6.3 Cross reference of classification. Part number M28787/76-1 supersedes part number M28787/76. The two parts are interchangeable.

*6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

| | | |
|---|---|---|
| Maximum power dissipation | | 1,030 mW (outputs not loaded) |
| Supply voltage | | 5 V dc ±10 percent |
| Input voltage | | |
| Low level | | -0.5 V minimum, 0.8 V maximum |
| High level | | 2.0 V minimum, $V_{CC}$ maximum |
| Input current, $\overline{D0}$ | | |
| Low Level | ($V_{IL}$ = 0.0 V) | -1.73 mA Maximum |
| | ($V_{IL}$ = 0.4 V) | -1.6 mA maximum |
| High Level | ($V_{IH}$ = 2.4 V) | 40 µA maximum |
| | ($V_{IH}$ = 5.5 V) | 1.0 mA maximum |
| Input current, $\overline{D1}$-$\overline{D7}$, $\overline{E1}$ | | |
| Low level | ($V_{IL}$ = 0.0 V) | -3.46 mA maximum |
| | ($V_{IL}$ = 0.4 V) | -3.2 mA maximum |
| High level | ($V_{IH}$ = 2.4 V) | 80 µA maximum |
| | ($V_{IH}$ = 5.5 V) | 1.0 mA maximum |
| Output voltage, $\overline{GS}$, $\overline{E0}$ | | |
| Low level | ($I_{OL}$ = 16 mA) | 0.4 V maximum |
| High level | ($I_{OH}$ = -800 µA) | 2.4 V minimum |
| Output voltage A0-A2 | | |
| Low level | ($I_{OL}$ = 16 mA) | 0.4 V maximum |
| High level | ($I_{OH}$ = -400 µA) | 2.4 V minimum |
| Output voltage, $\overline{A0}$-$\overline{A2}$ | | |
| Low level | ($I_{OL}$ = 14.4 mA) | 0.4 V maximum |
| High level | ($I_{OH}$ = -760 µA) | 2.4 V minimum |
| Output drive current, $\overline{GS}$, $\overline{E0}$, maximum capability | | |
| Low level | ($V_{OL}$ = 0.4 V) | 16 mA |
| High level | ($V_{OH}$ = 2.4 V) | -800 µA |
| Output drive current, A0-A2, maximum capability | | |
| Low level | ($V_{OL}$ = 0.4 V) | 16 mA |
| High level | ($V_{OH}$ = 2.4 V) | -400 µA |
| Output drive current, $\overline{A0}$-$\overline{A2}$, maximum capability | | |
| Low level | ($V_{OL}$ = 0.4 V) | 14.4 mA |
| High level | ($V_{OH}$ = 2.4 V) | -760 µA |



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6.5 Module function. This module contains two 8-input priority encoders which produce a binary weighted code of the highest order input. Each priority encoder accepts data from eight active LOW inputs and provides a binary representation on three active LOW outputs and three active HIGH outputs. A priority is assigned to each input so that when two or more inputs are simultaneously active, the input with the highest priority is represented on the output, with input line 7 ($\overline{1D7}$ or $\overline{2D7}$) having the highest priority A HIGH on the input enable ($\overline{E1}$) will force all outputs to the inactive state and allow new data to settle without producing erroneous information at the outputs. A group signal ($\overline{GS}$) and an output enable ($\overline{EO}$) are provided with the three data outputs. $\overline{GS}$ is active level LOW when any input is LOW; this indicates when any input is active. The $\overline{EO}$ is active level LOW when all inputs are HIGH. Using the output enable along with the input enable allows priority encoding of N input signals. Both $\overline{EO}$ and $\overline{GS}$ are inactive HIGH when the input enable is HIGH.

6.5.1 Input to output relationship. Each priority encoder has the input to output relationship of table II for positive logic.

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Table II. Input to output relationship for each encoder. 1/-3/

| Inputs | | | | | | | | | Outputs | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\overline{EI}$ | $\overline{D0}$ | $\overline{D1}$ | $\overline{D2}$ | $\overline{D3}$ | $\overline{D4}$ | $\overline{D5}$ | $\overline{D6}$ | $\overline{D7}$ | $\overline{GS}$ | $\overline{A0}$ | $\overline{AI}$ | A2 | A0 | A1 | A2 | $\overline{EO}$ |
| H | X | X | X | X | X | X | X | X | H | H | H | H | L | L | L | H |
| L | H | H | H | H | H | H | H | H | H | H | H | H | L | L | L | L |
| L | X | X | X | X | X | X | X | L | L | L | L | L | H | H | H | H |
| L | X | X | X | X | X | X | L | H | L | H | L | L | L | H | H | H |
| L | X | X | X | X | X | L | H | H | L | L | H | L | H | L | H | H |
| L | X | X | X | X | L | H | H | H | L | H | H | L | L | L | H | H |
| L | X | X | X | L | H | H | H | H | L | L | L | H | H | H | L | H |
| L | X | X | L | H | H | H | H | H | L | H | L | H | L | H | L | H |
| L | X | L | H | H | H | H | H | H | L | L | H | H | H | L | L | H |
| L | L | H | H | H | H | H | H | H | L | H | H | H | L | L | L | H |

1/ H = High level.
2/ L = Low level.
3/ X = Don't care condition.





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6.5 Module function. This module contains two 8-input priority encoders which produce a binary weighted code of the highest order input. Each priority encoder accepts data from eight active LOW inputs and provides a binary representation on three active LOW outputs and three active HIGH outputs. A priority is assigned to each input so that when two or more inputs are simultaneously active, the input with the highest priority is represented on the output, with input line 7 ($\overline{1D7}$ or $\overline{2D7}$) having the highest priority A HIGH on the input enable ($\overline{EI}$) will force all outputs to the inactive state and allow new data to settle without producing erroneous information at the outputs. A group signal ($\overline{GS}$) and an output enable ($\overline{EO}$) are provided with the three data outputs. $\overline{GS}$ is active level LOW when any input is LOW; this indicates when any input is active. The $\overline{EO}$ is active level LOW when all inputs are HIGH. Using the output enable along with the input enable allows priority encoding of N input signals. Both $\overline{EO}$ and $\overline{GS}$ are inactive HIGH when the input enable is HIGH.

6.5.1 Input to output relationship. Each priority encoder has the input to output relationship of table II for positive logic.

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Table II. Input to output relationship for each encoder. 1/-3/

| Inputs | | | | | | | | | Outputs | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\overline{E1}$ | $\overline{D0}$ | $\overline{D1}$ | $\overline{D2}$ | $\overline{D3}$ | $\overline{D4}$ | $\overline{D5}$ | $\overline{D6}$ | $\overline{D7}$ | $\overline{GS}$ | $\overline{A0}$ | $\overline{A1}$ | A2 | A0 | A1 | A2 | $\overline{E0}$ |
| H | X | X | X | X | X | X | X | X | H | H | H | H | L | L | L | H |
| L | H | H | H | H | H | H | H | H | H | H | H | H | L | L | L | L |
| L | X | X | X | X | X | X | X | L | L | L | L | L | H | H | H | H |
| L | X | X | X | X | X | X | L | H | L | H | L | L | L | H | H | H |
| L | X | X | X | X | X | L | H | H | L | L | H | L | H | L | H | H |
| L | X | X | X | X | L | H | H | H | L | H | H | L | L | L | H | H |
| L | X | X | X | L | H | H | H | H | L | L | L | H | H | H | L | H |
| L | X | X | L | H | H | H | H | H | L | H | L | H | L | H | L | H |
| L | X | L | H | H | H | H | H | H | L | L | H | H | H | L | L | H |
| L | L | H | H | H | H | H | H | H | L | H | H | H | L | L | L | H |

1/ H = High level.
2/ L = Low level.
3/ X = Don't care condition.





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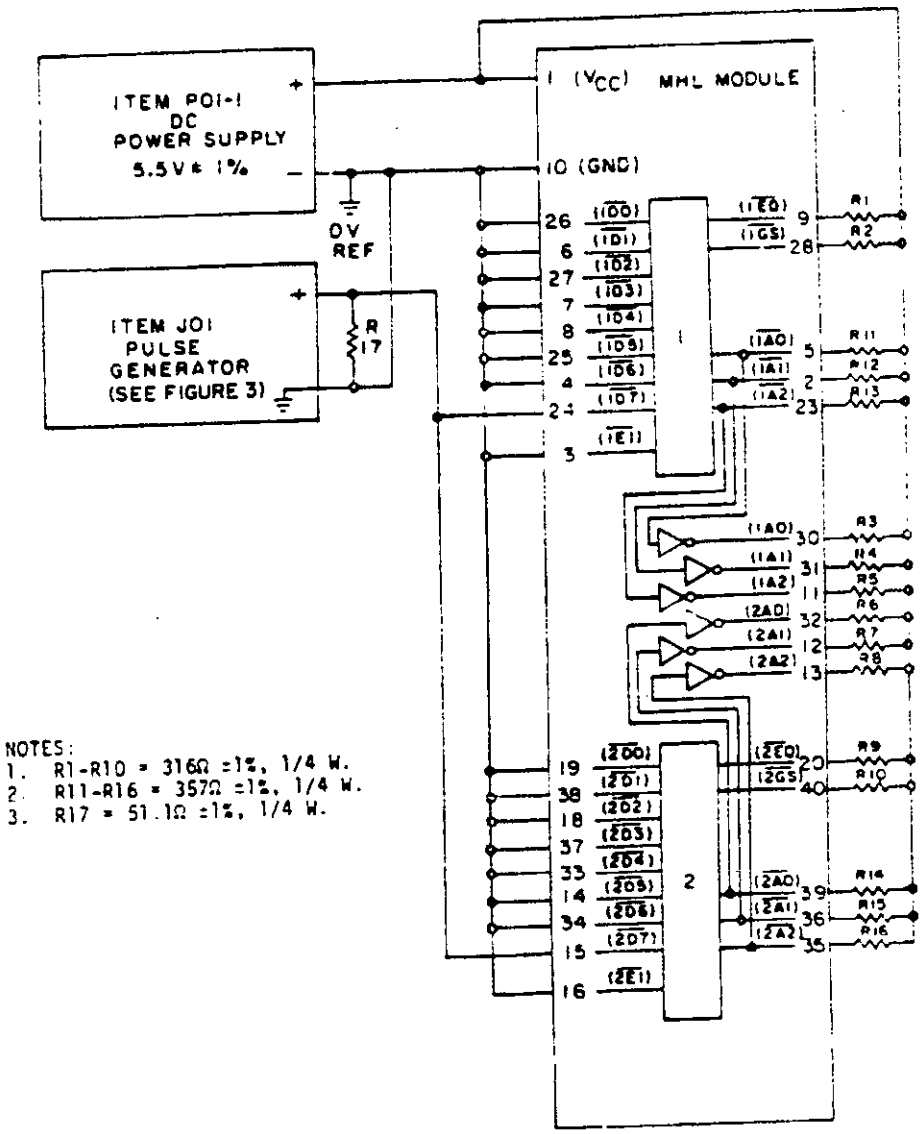
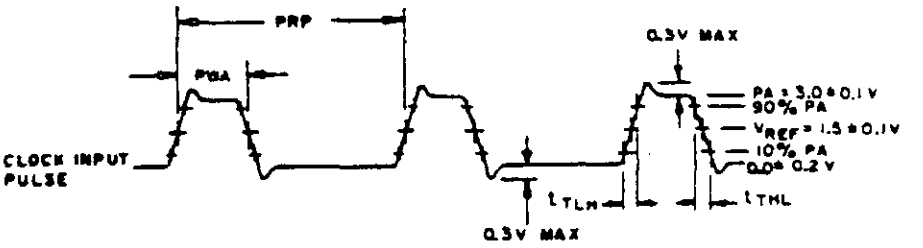



FIGURE 2. Operating temperature and life test circuit.



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NOTES:
1. Clock input pulse parameters:
   $t_{TLH} = t_{THL} = 10 \pm 2$ n..
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. Verify that all outputs have a high level $\geq$ 2.4 V and low level $\leq$ 0.4 V.
3. All voltages are referenced to ground.

FIGURE 3. Operating temperature and life test circuit waveform.



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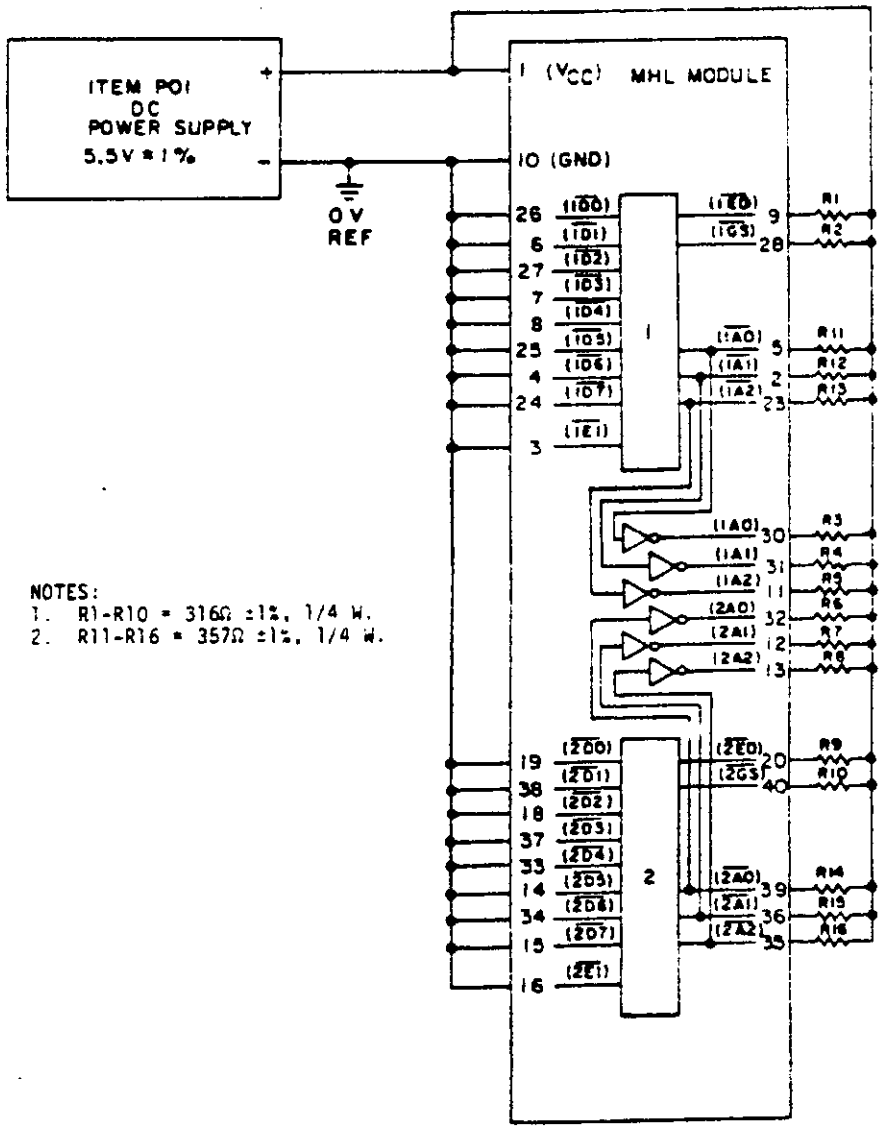



FIGURE 4. Component temperature test circuit.



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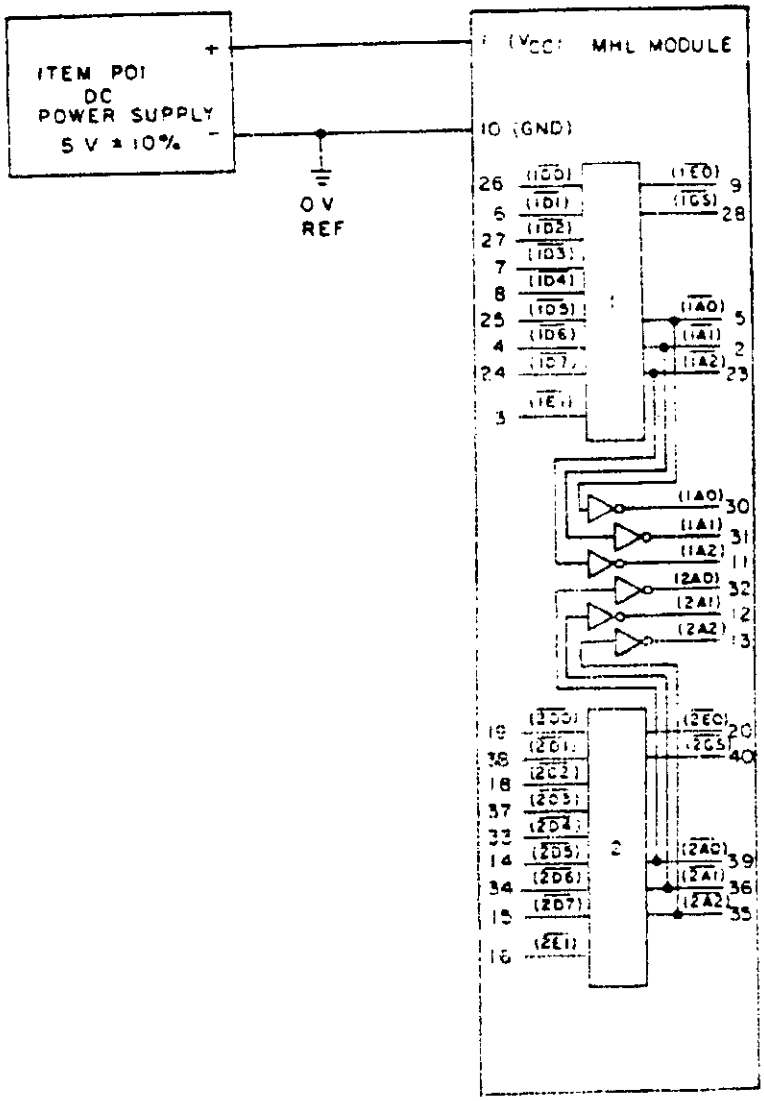



FIGURE 5. Vibration test circuit.



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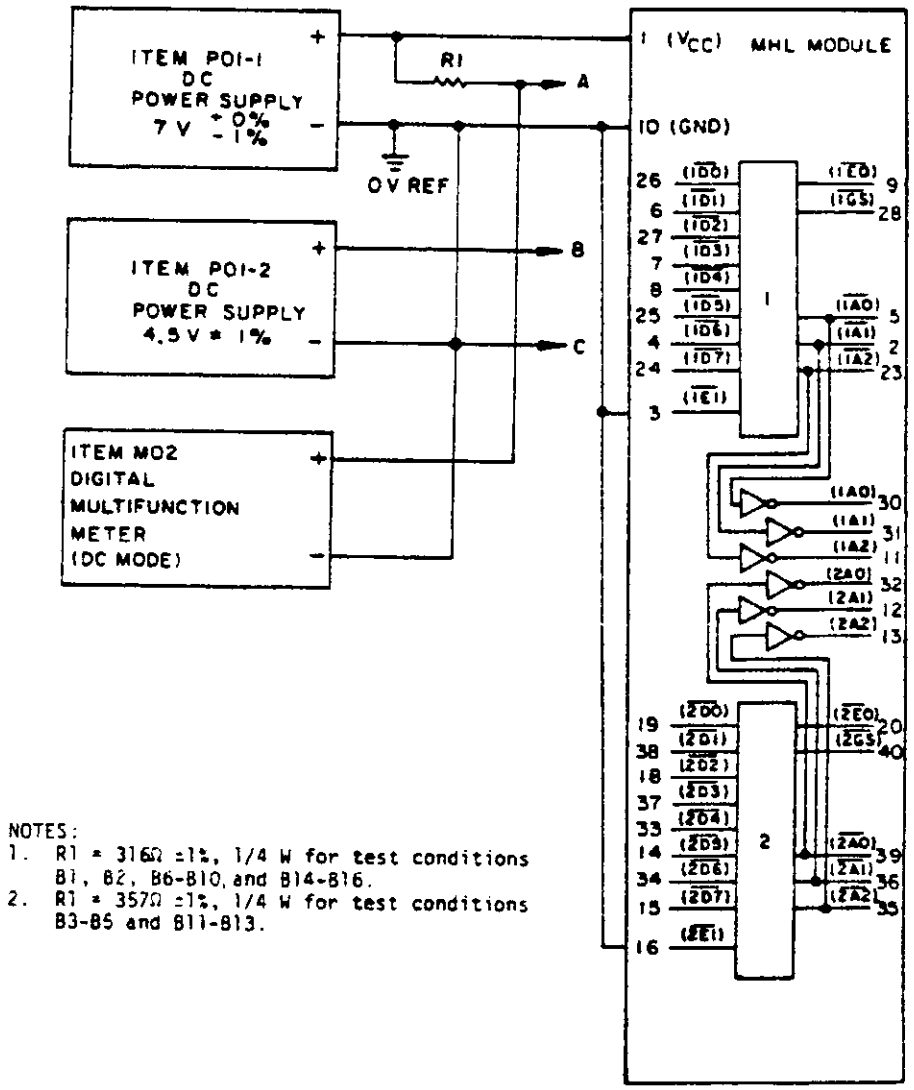
TABLE III. Isolation test.

| Test condition | Parameter | Connect | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| A1 | Isolation, frame | C,17,21,22,29 | Frame | 10 | ∞ | MΩ | 6 |
| A2 | Isolation, unused pins | C,21,22,29 | 17 | " | " | " | " |
| A3 | " | C,17,22,29 | 21 | " | " | " | " |
| A4 | " | C,17,21,29 | 22 | " | " | " | " |
| A5 | " | C,17,21,22 | 29 | " | " | " | " |





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* FIGURE 7. $V_{OUT}$ (low-loaded) test circuit for table IV.



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TABLE V. Input breakdown current test.

| Test condition | Parameter | Connect | | Limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| C1 | $\overline{IDO}$ Input breakdown current | 3,4,6,7,8,14,15,16,18,19,24,25,27,33,34,37,38 | 26 | 0 | 1 | mA | 8 |
| C2 | $\overline{ID1}$ " | 3,4,7,8,14,15,16,18,19,24,25,26,27,33,34,37,38 | 6 | " | " | " | " |
| C3 | $\overline{ID2}$ " | 3,4,6,7,8,14,15,16,18,19,24,25,26,33,34,37,38 | 27 | " | " | " | " |
| C4 | $\overline{ID3}$ " | 3,4,6,8,14,15,16,18,19,24,25,26,27,33,34,37,38 | 7 | " | " | " | " |
| C5 | $\overline{ID4}$ " | 3,4,6,7,14,15,16,18,19,24,25,26,27,33,34,37,38 | 8 | " | " | " | " |
| C6 | $\overline{ID5}$ " | 3,4,6,7,8,14,15,16,18,19,24,26,27,33,34,37,38 | 25 | " | " | " | " |
| C7 | $\overline{ID6}$ " | 3,6,7,8,14,15,16,18,19,24,25,26,27,33,34,37,38 | 4 | " | " | " | " |
| C8 | $\overline{ID7}$ " | 3,4,6,7,8,14,15,16,18,19,25,26,27,33,34,37,38 | 24 | " | " | " | " |
| C9 | $\overline{IET}$ " | 4,6,7,8,14,15,16,18,19,24,25,26,27,33,34,37,38 | 3 | " | " | " | " |





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TABLE V. Input breakdown current test - Continued.

| Test condition | Parameter | Connect | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| C10 | $\overline{2D0}$ Input breakdown current | 3,4,6,7,8,14, 15,16,18,24, 25,26,27,33, 34,37,38 | 19 | 0 | 1 | mA | 8 |
| C11 | $\overline{2D1}$ " | 3,4,6,7,8,14, 15,16,18,19, 24,25,26,27, 33,34,37 | 38 | " | " | " | " |
| C12 | $\overline{2D2}$ " | 3,4,6,7,8,14, 15,16,19,24, 25,26,27,33, 34,37,38 | 18 | " | " | " | " |
| C13 | $\overline{2D3}$ " | 3,4,6,7,8,14, 15,16,18,19, 24,25,26,27, 33,34,38 | 37 | " | " | " | " |
| C14 | $\overline{2D4}$ " | 3,4,6,7,8,14, 15,16,18,19, 24,25,26,27, 34,37,38 | 33 | " | " | " | " |
| C15 | $\overline{2D5}$ " | 3,4,6,7,8,15, 16,18,19,24, 25,26,27,33, 34,37,38 | 14 | " | " | " | " |
| C16 | $\overline{2D6}$ " | 3,4,6,7,8,14, 15,16,18,19, 24,25,26,27, 33,37,38 | 34 | " | " | " | " |
| C17 | $\overline{2D7}$ " | 3,4,6,7,8,14, 16,18,19,24, 25,26,27,33, 34,37,38 | 15 | " | " | " | " |
| C18 | $\overline{2F1}$ " | 3,4,6,7,8,14, 15,18,19,24, 25,26,27,33, 34,37,38 | 16 | " | " | " | " |





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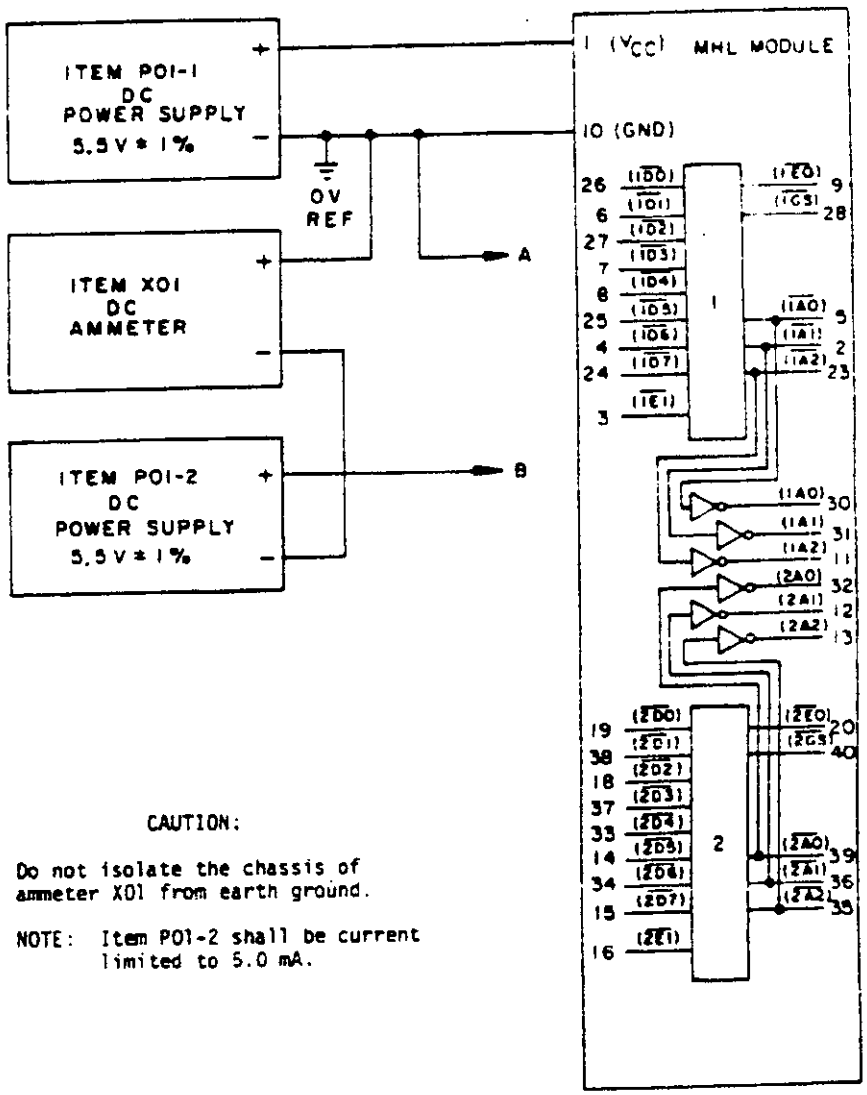



FIGURE 8. Input breakdown current test circuit for table V.



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TABLE VI. Output short circuit current test.

| Test condition | Parameter | Connect | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | Min | Max | | |
| D1 | $\overline{IGS}$ Output short circuit current | 3 | Open | 28 | -30 | -100 | mA | 9 |
| D2 | $\overline{1E0}$ " | " | " | 9 | " | " | " | " |
| D3 | $\overline{1A0}$ " | " | " | 5 | " | " | " | " |
| D4 | $\overline{1A1}$ " | " | " | 2 | " | " | " | " |
| D5 | $\overline{1A2}$ " | " | " | 23 | " | " | " | " |
| D6 | $\overline{2GS}$ " | 16 | " | 40 | " | " | " | " |
| D7 | $\overline{2E0}$ " | " | " | 20 | " | " | " | " |
| D8 | $\overline{2A0}$ " | " | " | 39 | " | " | " | " |
| D9 | $\overline{2A1}$ " | " | " | 36 | " | " | " | " |
| D10 | $\overline{2A2}$ " | " | " | 35 | " | " | " | " |
| D11 | 1A0 " | Open | 3 | 30 | -20 | -55 | " | " |
| D12 | 1A1 " | " | 3 | 31 | " | " | " | " |
| D13 | 1A2 " | " | 3 | 11 | " | " | " | " |
| D14 | 2A0 " | " | 16 | 32 | " | " | " | " |
| D15 | 2A1 " | " | 16 | 12 | " | " | " | " |
| D16 | 2A2 " | " | 16 | 13 | " | " | " | " |





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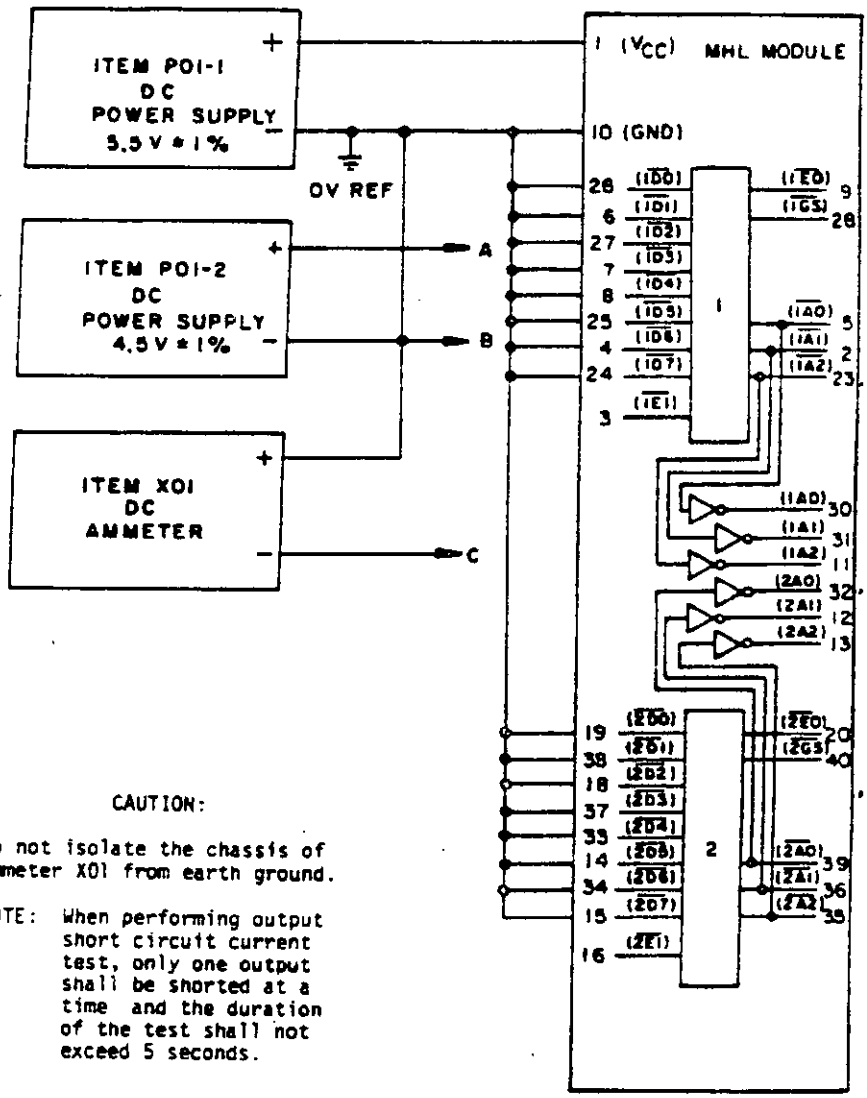



FIGURE 9. Output short circuit current test circuit for table VI.



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TABLE VII. Functional test.

| Test condition | Parameter | Connect | | | Output logic state | | | | | | | | | | | | | | | | Initial limits 25°C ± 5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | $\overline{IGS}$ | $\overline{IE0}$ | $\overline{IA0}$ | $\overline{IA1}$ | $\overline{IA2}$ | $\overline{I}_{A0}$ | $\overline{I}_{A1}$ | $\overline{I}_{A2}$ | $\overline{2GS}$ | $\overline{2E0}$ | $\overline{2A0}$ | $\overline{2A1}$ | $\overline{2A2}$ | $\overline{2}_{A0}$ | $\overline{2}_{A1}$ | $\overline{2}_{A2}$ | Min | Max | | |
| E1 | Functional | 4,6,8,14, 15,19,24, 25,26,27, 33,34,37, 38 | 3,7,16, 18 | 2/,3/ | L | H | L | L | H | H | H | L | L | H | H | L | H | L | H | L | | 1/ | | 10 |
| E2 | " | 4,7,8,14, 15,18,19, 24,26,27, 33,34,37, 38 | 3,6,16, 25 | " | L | H | L | H | L | H | L | H | H | L | H | H | H | L | L | L | | " | | " |
| E3 | " | 3,6,7,14, 15,18,19, 25,26,27, 33,34,37, | 4,8,16, 24,38 | " | H | H | H | H | H | L | L | L | L | H | L | H | H | H | L | L | | " | | " |
| E4 | " | 4,7,8,14, 16,24,25, 26,27,33, 34,38, | 3,6,15, 18,19, 37 | " | L | H | L | H | H | H | L | L | H | H | H | H | H | L | L | L | | " | | " |
| E5 | " | 4,7,8,16, 19,24,25, 38 | 3,6,14, 15,18, 26,27, 33,34, 37 | " | L | H | H | L | H | L | H | L | H | H | H | H | H | L | L | L | | " | | " |
| E6 | " | 4,14,15, 26,27,33, 34,37 | 3,6,7, 8,16, 18,19, 25,26, 38 | " | L | H | L | H | L | H | L | H | L | H | H | L | H | L | H | L | | " | | " |
| E7 | " | 3,4,6,7, 14,15,18, 26,33,34 37,38 | 8,16, 19,24, 25,27, | " | H | H | H | H | H | L | L | L | L | H | H | H | H | L | L | L | | " | | " |

See footnotes at end of table.





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TABLE VII. Functional test - Continued.

| Test condition | Parameter | Connect | | | Output logic states | | | | | | | | | | | | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | $\overline{T}_{GS}$ | $\overline{T}_{EO}$ | $\overline{T}_{A0}$ | $\overline{T}_{A1}$ | $\overline{T}_{A2}$ | $\overline{1}_{A0}$ | $\overline{1}_{A1}$ | $\overline{1}_{A2}$ | $\overline{2}_{GS}$ | $\overline{2}_{EO}$ | $\overline{2}_{A0}$ | $\overline{2}_{A1}$ | $\overline{2}_{A2}$ | $\overline{2}_{A0}$ | $\overline{2}_{A1}$ | $\overline{2}_{A2}$ | Min | Max | | |
| E8 | Functional | 4,6,7,8, 15,16,18, 19,24,25, 26,33,34, 37,38 | 3,14,27 | 2/,3/ | L | H | H | L | H | L | H | L | H | H | H | H | H | L | L | L | | 1/ | | 10 |
| E9 | " | 4,6,7,8, 24,25,26, 27,33,34, 38 | 3,14,15, 16,18, 19,37 | " | H | L | H | H | H | L | L | L | L | H | L | L | L | H | H | H | | " | | " |
| E10 | " | 3,4,8,14, 15,18,19, 24,33,34, 38 | 6,7,16, 25,26, 27,37 | " | H | H | H | H | H | L | L | L | L | H | L | L | H | H | H | L | | " | | " |
| E11 | " | 4,6,8,14, 15,18,19, 24,25,27, 33,37,38 | 3,7,16, 26,34 | " | L | H | L | L | H | H | H | L | L | H | H | L | L | L | L | H | | " | | " |
| E12 | " | 4,15,16, 24,25,26, 27,34,37 | 3,6,7, 8,14, 18,19, 33,38 | " | L | H | H | H | L | L | L | H | H | H | H | H | H | L | L | L | | " | | " |
| E13 | " | 3,6,7,14, 15,18,19, 26,34,37, 38 | 4,8,16, 24,25, 27,33 | " | H | H | H | H | H | L | L | L | L | H | H | H | L | L | L | H | | " | | " |
| E14 | " | 4,6,7,8, 14,19,24, 25,27,33, 34,37 | 3,15, 16,18, 26,38 | " | L | H | H | H | H | L | L | L | L | H | L | L | L | H | H | H | | " | | " |

See footnotes at end of table.





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TABLE VII. Functional test - Continued.

| Test condition | Parameter | Connect | | | Output logic states | | | | | | | | | | | | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | $\overline{TGS}$ | $\overline{TEO}$ | $\overline{TA0}$ | $\overline{TA1}$ | $\overline{TA2}$ | I1A0 | I1A1 | I1A2 | $\overline{ZGS}$ | $\overline{ZEO}$ | $\overline{ZA0}$ | $\overline{ZA1}$ | $\overline{ZA2}$ | I2A0 | I2A1 | I2A2 | Min | Max | | |
| E15 | Functional | 3,4,7,8, 14,15,19, 24,34,38 | 6,16, 18,25, 26,27, 33,37 | 2/,3/ | H | H | H | H | H | L | L | L | L | H | H | H | L | L | L | H | | 1/ | | 10 |
| E16 | " | 4,7,8,14, 16,18,24, 37 | 3,6,15, 19,25, 26,27, 33,34, 38 | " | L | H | L | H | L | H | L | H | H | H | H | H | H | L | L | L | | " | | " |
| E17 | " | 3,4,7,15, 18,19,24, 26,27,33, 34,38 | 6,8,14, 16,25, 37 | " | H | H | H | H | H | L | L | L | L | H | L | H | L | H | L | H | | " | | " |
| E18 | " | 3,4,6,7, 8,15,19, 24,25,26, 27,33,34, 37 | 14,16, 18,38 | " | H | H | H | H | H | L | L | L | L | H | L | H | L | H | L | H | | " | | " |
| E19 | " | 4,6,7,8, 15,16,18, 24,26,27, 34,37,38 | 3,14, 19,25, 33 | " | L | H | L | H | L | H | L | H | H | H | H | H | H | L | L | L | | " | | " |
| E20 | " | 4,8,16, 19,24,26 27,33 | 3,6,7, 14,15, 18,25, 34,37, 38 | " | L | H | L | H | L | H | L | H | H | H | H | H | H | L | L | L | | " | | " |
| E21 | " | 3,4,7,8, 16,19,26 27,33 | 6,14, 15,18, 24,25, 34,37, 38 | " | H | H | H | H | H | L | L | L | H | H | H | H | H | L | L | L | | " | | " |

See footnotes at end of table.





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TABLE VII. Functional test - Continued.

| Test condition | Parameter | Connect | | | Output logic states | | | | | | | | | | | | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | $\overline{T}_{GS}$ | $\overline{T}_{EO}$ | $\overline{T}_{A0}$ | $\overline{T}_{A1}$ | $\overline{T}_{A2}$ | $\overline{I}_{A0}$ | $\overline{I}_{A1}$ | $\overline{I}_{A2}$ | $\overline{Z}_{GS}$ | $\overline{Z}_{EO}$ | $\overline{Z}_{A0}$ | $\overline{I}_{A0}$ | $\overline{Z}_{A1}$ | $\overline{Z}_{A2}$ | $Z_{A0}$ | $Z_{A1}$ | $Z_{A2}$ | Min | Max | | |
| E22 | Functional | 6,7,15, 18,19,24, 25,33,34, 37,38 | 3,4,8, 14,16, 26,27, | 2/,3/ | L | H | H | L | L | L | H | H | L | H | L | H | L | H | L | H | | 1/ | | 10 |
| E23 | " | 3,7,8,14, 15,16,18, 19,33,34, 37,38 | 4,6,24, 25,26, 27 | " | H | H | H | H | H | L | L | L | H | H | H | H | H | L | L | L | | " | | " |
| E24 | " | 3,4,8,14, 15,16,18, 25,26,33, 34,38 | 6,7,19, 24,27, 37 | " | H | H | H | H | H | L | L | L | H | H | H | H | H | L | L | L | | " | | " |
| E25 | " | 7,14,15, 18,34 | 3,4,6, 8,16, 19,24, 25,26, 27,33, 37,38 | " | L | H | L | L | L | H | H | H | L | H | H | H | L | L | L | H | | " | | " |
| E26 | " | 4,6,7,14, 18,24,25, 26,27,34, 38 | 3,8,15, 16,19, 33,37, | " | L | H | H | H | L | L | L | H | L | H | L | L | L | H | H | H | | " | | " |
| E27 | " | 4,6,7,18, 24,25, | 3,8,14, 15,16, 19,26, 27,33, 34,37, 38 | " | L | H | H | H | L | L | L | H | L | H | L | L | L | H | H | H | | " | | " |
| E28 | " | 6,7,15, 18,19,24, 25,27,33 | 3,4,8, 14,16, 26,34, 37,38, | " | L | H | H | L | L | L | H | H | L | H | H | L | L | L | H | H | | " | | " |

See footnotes at end of table.





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TABLE VII. Functional test - Continued.

| Test condition | Parameter | Connect | | | Output logic states | | | | | | | | | | | | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | $\overline{T}_{GS}$ | $\overline{T}_{E0}$ | $T_{A0}$ | $T_{A1}$ | $T_{A2}$ | $I_{A0}$ | $I_{A1}$ | $I_{A2}$ | $\overline{Z}_{GS}$ | $\overline{Z}_{E0}$ | $\overline{Z}_{A0}$ | $\overline{Z}_{A1}$ | $\overline{Z}_{A2}$ | $I2_{A0}$ | $I2_{A1}$ | $I2_{A2}$ | Min | Max | | |
| E29 | Functional | 7,14,16, 18,24,25, 27,33,34, 37 | 3,4,6, 8,15, 19,26, 38 | 2/,3/ | L | H | H | L | L | L | H | H | H | H | H | H | H | L | L | L | | 1/ | | 10 |
| E30 | " | 3,4,6,14, 18,19,25, 33,34,37, 38 | 7,8,15, 16,24, 26,27 | " | H | H | H | H | H | L | L | L | L | H | L | L | L | H | H | H | | " | | " |
| E31 | " | 8,18,19, 24,25,26, 27,37,38 | 3,4,6, 7,14, 15,16, 33,34 | " | L | H | H | L | L | L | H | H | L | H | L | L | L | H | H | H | | " | | " |
| E32 | " | 4,7,14, 15,18,19, 33,38 | 3,6,8, 16,24, 25,26, 27,34, 37 | " | L | H | L | L | L | H | H | H | L | H | H | L | L | L | H | H | | " | | " |
| E33 | " | 6,7,8,19, 24,25,26, 27,33,37, 38 | 3,4,14, 15,16, 18,34 | " | L | H | H | L | L | L | H | H | L | H | L | L | L | H | H | H | | " | | " |
| E34 | " | 6,18,19, 24,33,37, | 3,4,7, 8,14, 15,16, 25,26, 27,34, 36 | " | L | H | H | L | L | L | H | H | L | H | L | L | L | H | H | H | | " | | " |

See footnotes at end of table.





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TABLE VII. Functional test - Continued.

| Test condition | Parameter | Connect | | | Output logic states | | | | | | | | | | | | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | $\overline{1GS}$ | $\overline{1E0}$ | $\overline{1A0}$ | $\overline{1A1}$ | $\overline{1A2}$ | 1A0 | 1A1 | 1A2 | $\overline{2GS}$ | $\overline{2E0}$ | $\overline{2A0}$ | $\overline{2A1}$ | $\overline{2A2}$ | 2A0 | 2A1 | 2A2 | Min | Max | | |
| E35 | Functional | 4,6,7,8, 19,25,26, 27,34,38, | 3,14, 15,16, 18,24, 33,37 | 2/,3/ | L | H | L | L | L | H | H | H | L | H | L | L | L | H | H | H | 1/ | | | 10 |

1/ Output logic states:
L = Low-voltage level = 0.0 to 0.4 V.
H = High-voltage level = 2.4 to 4.5 V.

2/ For test conditions E1-E35 verification of the output logic states in 1/ is obtained by connecting item M02, point C, to each output pin in succession after input conditions have been accomplished.

3/ $\overline{1GS}$, $\overline{1E0}$, $\overline{1A0}$, $\overline{1A1}$, $\overline{1A2}$, 1A0, 1A1, 1A2, $\overline{2GS}$, $\overline{2E0}$, $\overline{2A0}$, $\overline{2A1}$, $\overline{2A2}$, 2A0, 2A1, and 2A2 output logic states are measured at output pins 28, 9, 5, 2, 23, 30, 31, 11, 40, 20, 39, 36, 35, 32, 12, and 13 respectively.





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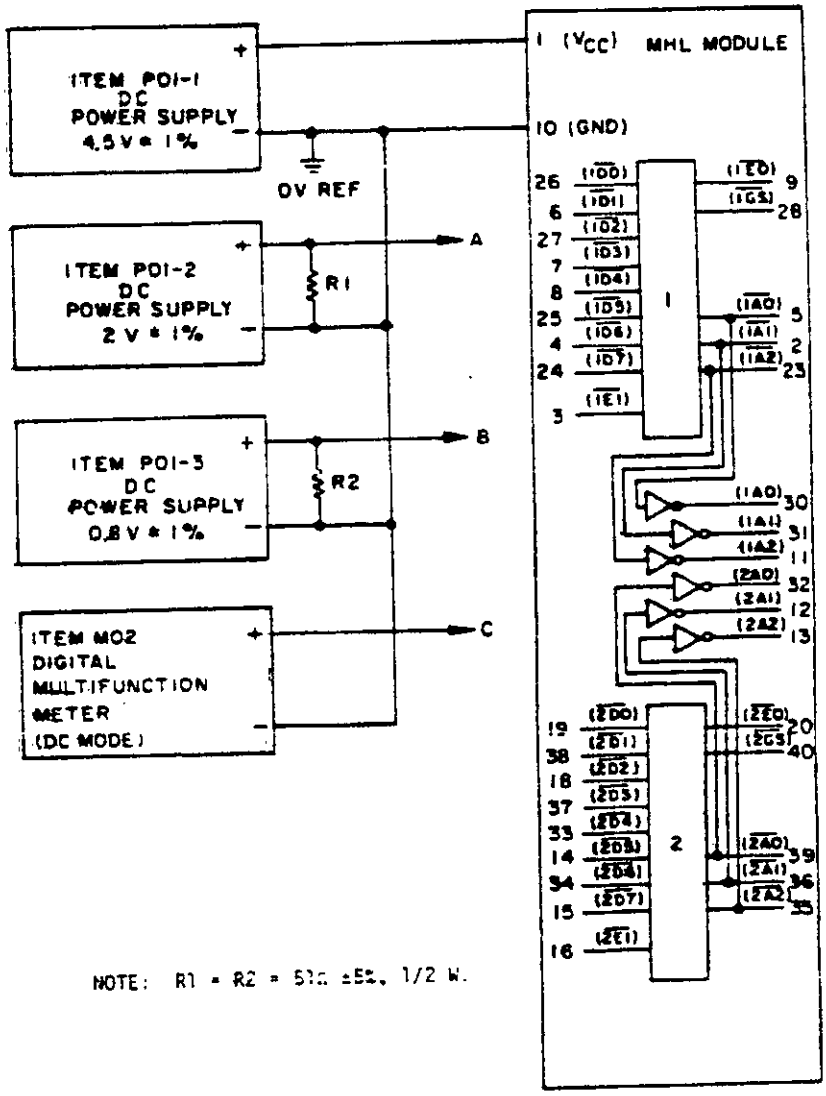



FIGURE 10. Functional test circuit for table VII.



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TABLE VIII. Input clamp diode voltage test.

| Test condition | Parameter | Connect | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|
| | | A to pin | Min | Max | | |
| F1 | $\overline{IEI}$ Input clamp diode voltage | 3 | -0.4 | -1.5 | V | 11 |
| F2 | $\overline{I00}$ " | 26 | " | " | " | " |
| F3 | $\overline{I01}$ " | 6 | " | " | " | " |
| F4 | $\overline{I02}$ " | 27 | " | " | " | " |
| F5 | $\overline{I03}$ " | 7 | " | " | " | " |
| F6 | $\overline{I04}$ " | 8 | " | " | " | " |
| F7 | $\overline{I05}$ " | 25 | " | " | " | " |
| F8 | $\overline{I06}$ " | 4 | " | " | " | " |
| F9 | $\overline{I07}$ " | 24 | " | " | " | " |
| F10 | $\overline{2EI}$ " | 16 | " | " | " | " |
| F11 | $\overline{200}$ " | 19 | " | " | " | " |
| F12 | $\overline{201}$ " | 38 | " | " | " | " |
| F13 | $\overline{202}$ " | 18 | " | " | " | " |
| F14 | $\overline{203}$ " | 37 | " | " | " | " |
| F15 | $\overline{204}$ " | 33 | " | " | " | " |
| F16 | $\overline{205}$ " | 14 | " | " | " | " |
| F17 | $\overline{206}$ " | 34 | " | " | " | " |
| F18 | $\overline{207}$ " | 15 | " | " | " | " |





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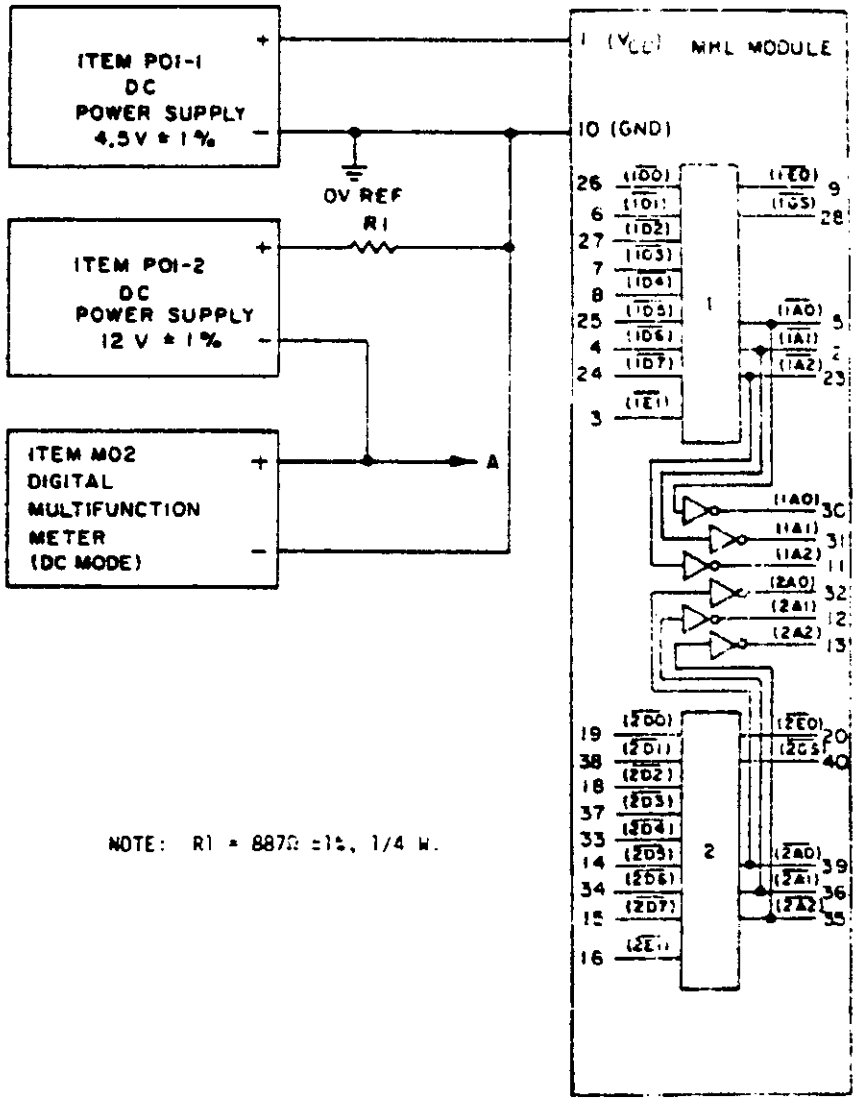



FIGURE 11. Input clamp diode voltage test circuit for table VIII.



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TABLE IX. Output voltage (low-level) test.

| Test condition | Parameter | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| G1 | $\overline{TGS}$ Output voltage, low level | Open | 24 | 28 | Open | 0.0 | 0.4 | V | 12 |
| G2 | $\overline{TA0}$ " | " | " | 5 | " | " | " | " | " |
| G3 | $\overline{TA1}$ " | " | " | 2 | " | " | " | " | " |
| G4 | $\overline{TA2}$ " | " | " | 23 | " | " | " | " | " |
| G5 | TA0 " | 4,6,7,8,24, 25,26,27 | Open | 30 | 5 | " | " | " | " |
| G6 | TA1 " | " | " | 31 | 2 | " | " | " | " |
| G7 | TA2 " | " | " | 11 | 23 | " | " | " | " |
| G8 | $\overline{TE0}$ " | " | " | 9 | Open | " | " | " | " |
| G9 | $\overline{2GS}$ " | Open | 15 | 40 | " | " | " | " | " |
| G10 | $\overline{2A0}$ " | " | " | 39 | " | " | " | " | " |
| G11 | $\overline{2A1}$ " | " | " | 36 | " | " | " | " | " |
| G12 | $\overline{2A2}$ " | " | " | 35 | " | " | " | " | " |
| G13 | 2A0 " | 14,15,18, 19,33,34, 37,38 | Open | 32 | 19 | " | " | " | " |
| G14 | 2A1 " | " | " | 12 | 36 | " | " | " | " |
| G15 | 2A2 " | " | " | 13 | 35 | " | " | " | " |
| G16 | $\overline{2E0}$ " | " | " | 20 | Open | " | " | " | " |





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TABLE X. Output voltage (high-level) test.

| Test condition | Parameter | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test Figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| H1 | $\overline{TGS}$ Output voltage, high level | 4,6,7,8,14, 15,18,19, 24,25,26, 27,33,34, 37,38 | Open | 28 | Open | 2.4 | 4.5 | V | 12 |
| H2 | $\overline{TGS}$ " | 4,6,7,8,14, 18,19,24, 25,26,27, 33,34,37, 38 | 15 | 28 | " | " | " | " | " |
| H3 | $\overline{TAU}$ " | 4,6,7,8,14, 15,18,19, 24,25,26, 27,33,34, 37,38 | Open | 5 | " | " | " | " | " |
| H4 | $\overline{TAU}$ " | 6,7,8,14, 18,19,24, 25,26,27, 33,34,37, 38 | 4,15 | 5 | " | " | " | " | " |
| H5 | $\overline{TAT}$ " | 4,6,7,8,14, 15,18,19, 24,25,26, 27,33,34, 37,38 | Open | 2 | " | " | " | " | " |
| H6 | $\overline{TAT}$ " | 4,6,7,8,14, 18,19,24, 26,27,33, 34,37,38, | 15,25 | 2 | " | " | " | " | " |
| H7 | $\overline{TA2}$ " | 4,6,7,8,14, 15,18,19, 24,25,26, 27,33,34, 37,38 | Open | 23 | " | " | " | " | " |





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TABLE X. Output voltage (high-level) test - Continued.

| Test condition | Parameter | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| H8 | IA2 Output voltage, high level | 4,6,8,14, 18,19,24, 25,26,27, 33,34,37, 38 | 7,15 | 23 | Open | 2.4 | 4.5 | V | 12 |
| H9 | IA0 " | 4,7,8,14, 15,18,19, 24,25,26, 27,33,34, 37,38 | 5 | 30 | 6 | " | " | " | " |
| H10 | IA0 " | 4,5,7,8,14, 18,19,25, 26,27,33, 34,37,38 | 15,24 | 31 | 2 | " | " | " | " |
| H11 | IA1 " | 4,6,7,8,14, 15,18,19, 24,25,26, 33,34,37, 38 | 27 | 31 | 2 | " | " | " | " |
| H12 | IA1 " | 4,6,7,8,14, 18,19,25, 26,27,33, 34,37,38, | 15,24 | 31 | 2 | " | " | " | " |
| H13 | IA2 " | 4,6,7,14, 15,18,19, 24,25,26, 27,33,34, 37,38 | 9 | 11 | 23 | " | " | " | " |
| H14 | IA2 " | 4,6,7,8,14, 18,19,25, 26,27,33, 34,37,38 | 15,24 | 11 | 23 | " | " | " | " |





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TABLE X. Output voltage (high-level) test - Continued.

| Test condition | Parameter | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| H15 | $\overline{TEU}$ Output voltage, high level | 4,6,7,8,14, 15,18,19, 25,26,27, 33,34,37, 38 | 24 | 9 | Open | 2.4 | 6.5 | V | 12 |
| H16 | $\overline{TEU}$ " | 4,6,7,8,14, 18,19,24, 25,27,33, 34,37,38 | 15,26 | 9 | " | " | " | " | " |
| H17 | $\overline{2GS}$ " | 4,6,7,8,14, 15,18,19, 24,25,26, 27,33,34, 37,38 | Open | 40 | " | " | " | " | " |
| H18 | $\overline{2GS}$ " | 4,6,7,8,14, 15,18,19, 25,26,27, 33,34,37, 38 | 24 | 40 | " | " | " | " | " |
| H19 | $\overline{2AO}$ " | 4,6,7,8,14, 15,18,19, 24,25,26, 27,33,34, 37,38 | Open | 39 | " | " | " | " | " |
| H20 | $\overline{2AO}$ " | 4,6,7,8,14, 15,18,19, 25,26,27, 33,37,38 | 24,34 | 39 | " | " | " | " | " |
| H21 | $\overline{2AT}$ " | 4,6,7,8,14, 15,18,19, 24,25,26, 27,33,34, 37,38 | Open | 36 | " | " | " | " | " |





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TABLE X. Output voltage (high-level) test - Continued.

| Test condition | Parameter | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| H22 | $\overline{2A1}$ Output voltage, high level | 4,6,7,8,15, 18,19,25, 26,27,33, 34,37,38 | 14,24 | 36 | Open | 2.4 | 4.5 | V | 12 |
| H23 | $\overline{2A2}$ " | 4,6,7,8,14, 15,18,19, 24,25,26, 27,33,34, 37,38 | Open | 35 | Open | " | " | " | " |
| H24 | $\overline{2A2}$ " | 4,6,7,8,14, 15,18,19, 25,26,27, 33,34,38 | 24,37 | 35 | Open | " | " | " | " |
| H25 | 2A0 " | 4,6,7,8,14, 15,18,19, 24,25,26, 27,33,34, 37 | 38 | 32 | 39 | " | " | " | " |
| H26 | 2A0 " | 4,6,7,8,14, 18,19,25, 26,27,33, 34,37,38 | 15,24 | 32 | 39 | " | " | " | " |
| H27 | 2A1 " | 4,6,7,8,14, 15,19,24, 25,26,27, 33,34,37, 38 | 18 | 12 | 36 | " | " | " | " |
| H28 | 2A1 " | 4,6,7,8,14, 18,19,25, 26,27,33, 34,37,38 | 15,24 | 12 | 36 | " | " | " | " |





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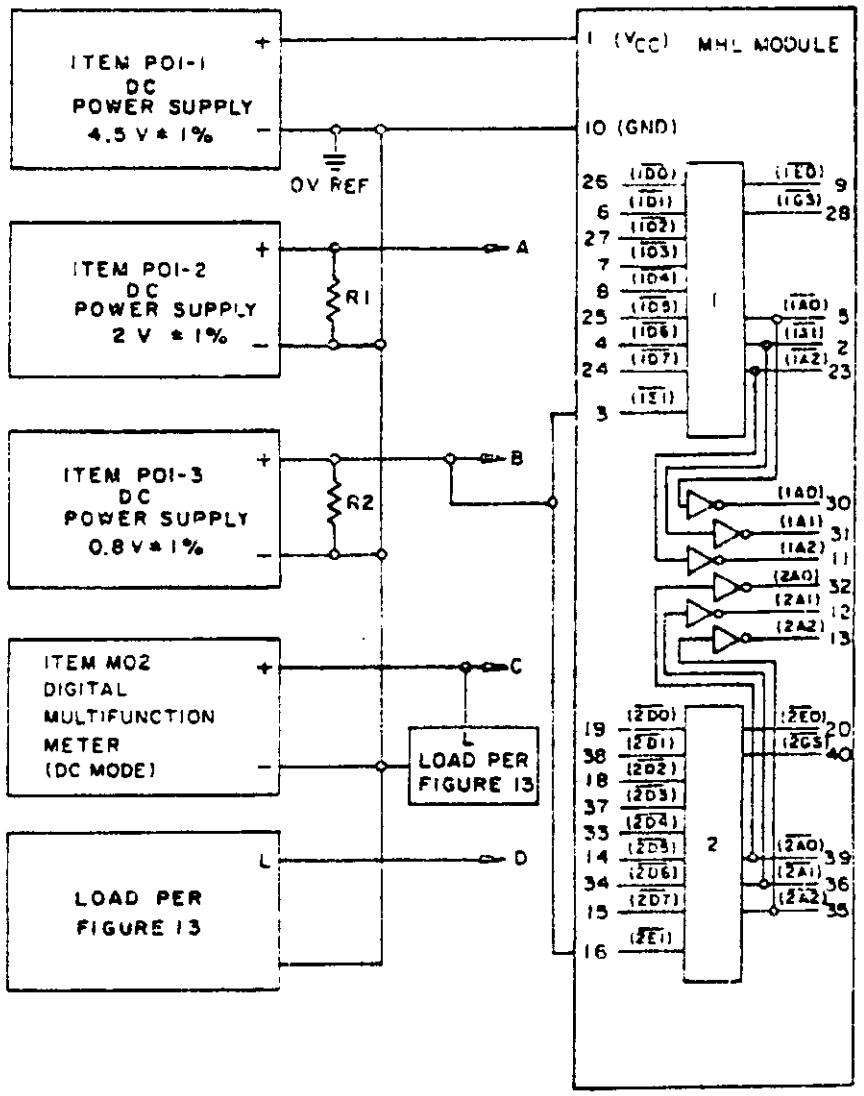
TABLE X. Output voltage (high-level) test - Continued.

| Test condition | Parameter | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| H29 | 2A2 Output voltage, high level | 4,6,7,8,14, 15,18,19, 24,25,26, 27,34,37, 38 | 33 | 13 | 35 | 2.4 | 4.5 | V | 12 |
| H30 | 2A2 " | 4,6,7,8,14, 18,19,25, 26,27,33, 34,37,38 | 15,24 | 13 | 35 | " | " | " | " |
| H31 | $\overline{2E0}$ " | 4,6,7,8,14, 18,19,24, 25,26,27, 33,34,37, 38 | 15 | 20 | Open | " | " | " | " |
| H32 | $\overline{2E0}$ " | 4,6,7,8,14, 15,18,25, 26,27,33, 34,37,38, | 19,24 | 20 | Open | " | " | " | " |





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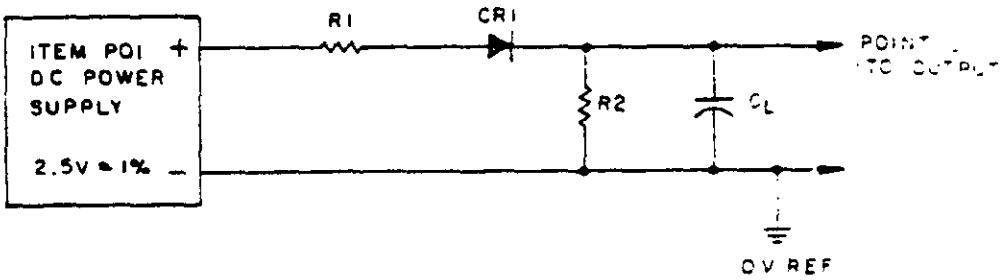



NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

FIGURE 12. Output voltage (low-level and high-level) test circuit for tables IX and X.



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NOTES:

1. The configuration shown above is the same for all load circuits and the following notes are the same for all load circuits:

   a. CR1 = 1N3600, 1N4150, or equivalent.
   b. Test type S:

      $C_L$ = 50 ±5 pF (including probe and parasitic capacitance).

      Test type Q:

      $C_L$ = 150 ±5 pF (including probe and parasitic capacitance).

   c. Capacitor $C_L$ is not required for test conditions G1-G16 and H1-H32.
   d. The lead length from the load circuit to the module output pin under test shall be less than 2 inches (50.8 mm).

2. Load circuit 1 for outputs $\overline{GS}$ and $\overline{ED}$:

   a. R1 = Select value for a current flow of 16 mA ±5% with point L held at 0.4 V.
   b. R2 = 2.94 kΩ ±1%, 1/8 W.

3. Load circuit 2 for outputs $\overline{A0}$-$\overline{A2}$:

   a. R1 = Select value for a current flow of 14.4 mA ±5% with point L held at 0.4 V.
   b. R2 = 3.16 kΩ ±1%, 1/8 W.

4. Load circuit 3 for outputs A0-A2:

   a. R1 = Select value for a current flow of 16 mA ±5% with point L held at 0.4 V.
   b. R2 = 6.04 kΩ ±1%, 1/8 W.

FIGURE 13. MHL logic load circuit.



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TABLE XI. Input current (high-level) test.

| Test condition | Parameter | Connect | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| J1 | $\overline{IO0}$ Input current, high level | 3,4,6,7,8,14,15,16,18, 19,24,25,27,33,34,37,38 | 26 | 0 | 40 | uA | 14 |
| J2 | $\overline{2O0}$ " | 3,4,6,7,8,14,15,16,18, 24,25,26,27,33,34,37,38 | 19 | " | 40 | " | " |
| J3 | $\overline{IO1}$ " | 3,4,7,8,14,15,16,18,19, 24,25,26,27,33,34,37,38 | 6 | " | 80 | " | " |
| J4 | $\overline{IO2}$ " | 3,4,6,7,8,14,15,16,18, 19,24,25,26,33,34,37,38 | 27 | " | " | " | " |
| J5 | $\overline{IO3}$ " | 3,4,6,8,14,15,16,18,19, 24,25,26,27,33,34,37,38 | 7 | " | " | " | " |
| J6 | $\overline{IO4}$ " | 3,4,6,7,14,15,16,18,19, 24,25,26,27,33,34,37,38 | 8 | " | " | " | " |
| J7 | $\overline{IO5}$ " | 3,4,6,7,8,14,15,16,18, 19,24,26,27,33,34,37,38 | 25 | " | " | " | " |
| J8 | $\overline{IO6}$ " | 3,6,7,8,14,15,16,18,19, 24,25,26,27,33,34,37,38 | 4 | " | " | " | " |
| J9 | $\overline{IO7}$ " | 3,4,6,7,8,14,15,16,18, 19,25,26,27,33,34,37,38 | 24 | " | " | " | " |
| J10 | $\overline{IE1}$ " | 4,6,7,8,14,15,16,18,19, 24,25,26,27,33,34,37,38 | 3 | " | " | " | " |
| J11 | $\overline{2O1}$ " | 3,4,6,7,8,14,15,16,18, 19,24,25,26,27,33,34,37 | 38 | " | " | " | " |
| J12 | $\overline{2O2}$ " | 3,4,6,7,8,14,15,16,19, 24,25,26,27,33,34,37,38 | 18 | " | " | " | " |
| J13 | $\overline{2O3}$ " | 3,4,6,7,8,14,15,16,18, 19,24,25,26,27,33,34,36 | 37 | " | " | " | " |
| J14 | $\overline{2O4}$ " | 3,4,6,7,8,14,15,16,18, 19,24,25,26,27,34,37,38 | 33 | " | " | " | " |





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TABLE XI. Input current (high-level) test - Continued.

| Test condition | Parameter | Connect | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| J15 | $\overline{205}$ Input current, high level | 3,4,6,7,8,15,16,18,19, 24,25,26,27,33,34,37,38 | 14 | 0 | 80 | µA | 14 |
| J16 | $\overline{206}$ " | 3,4,6,7,8,14,15,16,18, 19,24,25,26,27,33,37,38 | 34 | " | " | " | " |
| J17 | $\overline{207}$ " | 3,4,6,7,8,14,16,18, 19,24,25,26,27,33,34,37, 38 | 15 | " | " | " | " |
| J18 | $\overline{2E1}$ " | 3,4,6,7,8,14,15,18,19 24,25,26,27,33,34,37,38 | 16 | " | " | " | " |





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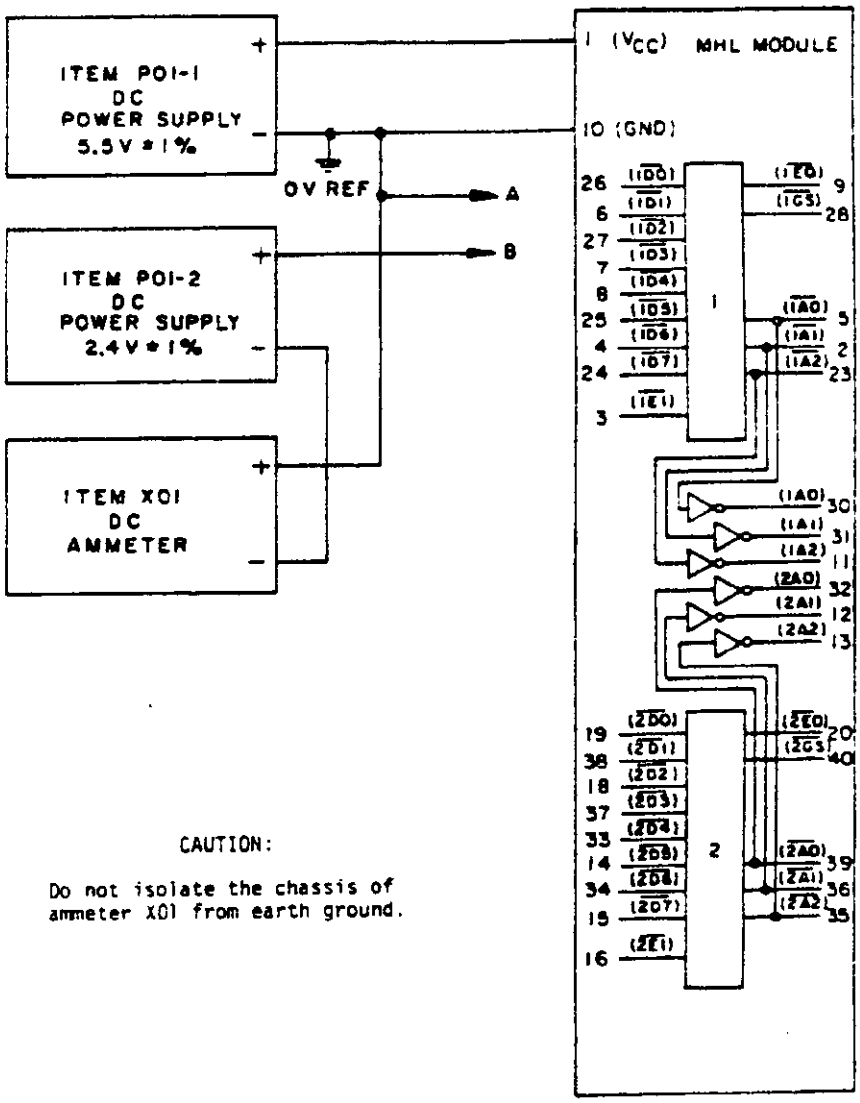



FIGURE 14. Input current (high-level) test circuit for table XI.



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TABLE XII. Input current (low-level) test.

| Test condition | Parameter | Connect | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|
| | | A to pin | Min | Max | | |
| K1 | $\overline{1D0}$ Input current, low level | 26 | -0.10 | -1.73 | mA | 15 |
| K2 | $\overline{2D0}$ " | 19 | -0.10 | -1.73 | " | " |
| K3 | $\overline{1D1}$ " | 6 | -0.20 | -3.46 | " | " |
| K4 | $\overline{1D2}$ " | 27 | " | " | " | " |
| K5 | $\overline{1D3}$ " | 7 | " | " | " | " |
| K6 | $\overline{1D4}$ " | 8 | " | " | " | " |
| K7 | $\overline{1D5}$ " | 25 | " | " | " | " |
| K8 | $\overline{1D6}$ " | 4 | " | " | " | " |
| K9 | $\overline{1D7}$ " | 24 | " | " | " | " |
| K10 | $\overline{1E1}$ " | 3 | " | " | " | " |
| K11 | $\overline{2D1}$ " | 38 | " | " | " | " |
| K12 | $\overline{2D2}$ " | 18 | " | " | " | " |
| K13 | $\overline{2D3}$ " | 37 | " | " | " | " |
| K14 | $\overline{2D4}$ " | 33 | " | " | " | " |
| K15 | $\overline{2D5}$ " | 14 | " | " | " | " |
| K16 | $\overline{2D6}$ " | 34 | " | " | " | " |
| K17 | $\overline{2D7}$ " | 15 | " | " | " | " |
| K18 | $\overline{2E1}$ " | 16 | " | " | " | " |





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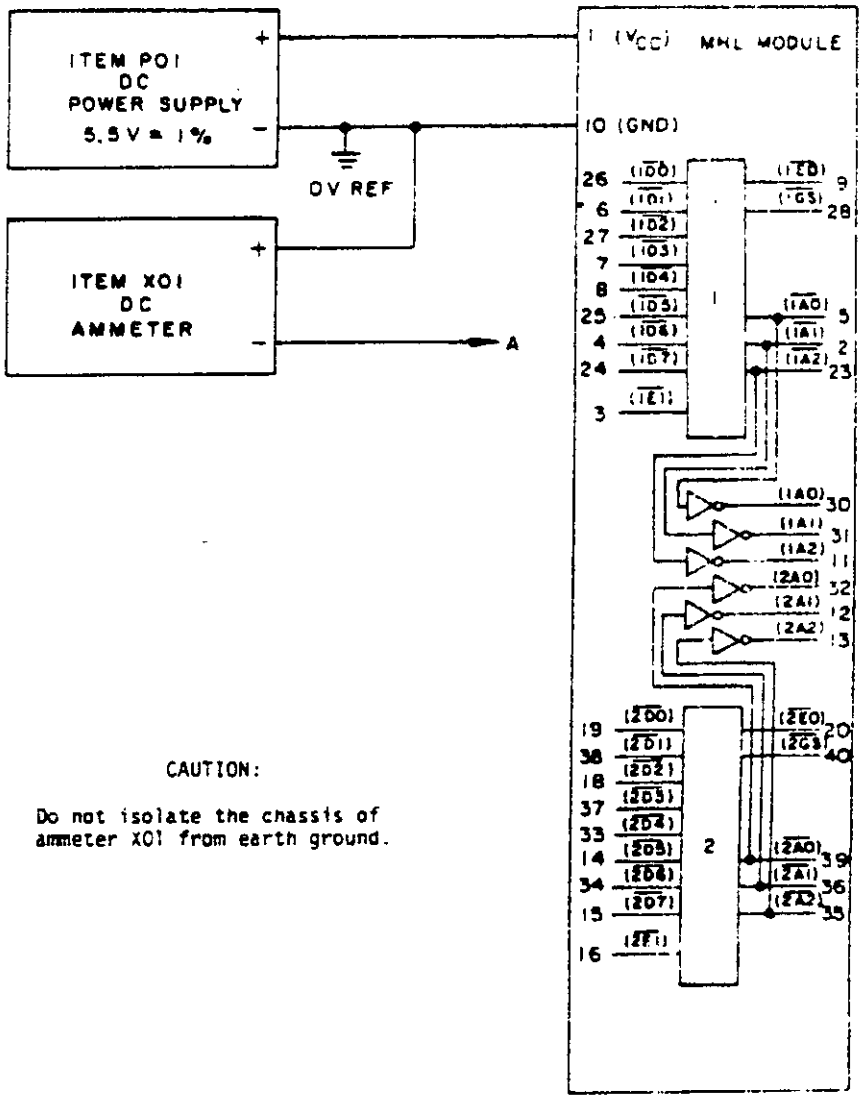



FIGURE 15. Input current (low-level) test circuit for table XII.



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TABLE XIII. Power supply current (low-level outputs and high-level outputs) test.

| Test condition | Parameter | Connect | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|
| | | A to pin | Min | Max | | |
| L1 | Power supply current, low-level outputs | Open | 77 | 187 | mA | 16 |
| L2 | Power supply current, high-level outputs | 3,15,16,24 | 77 | 187 | mA | 16 |





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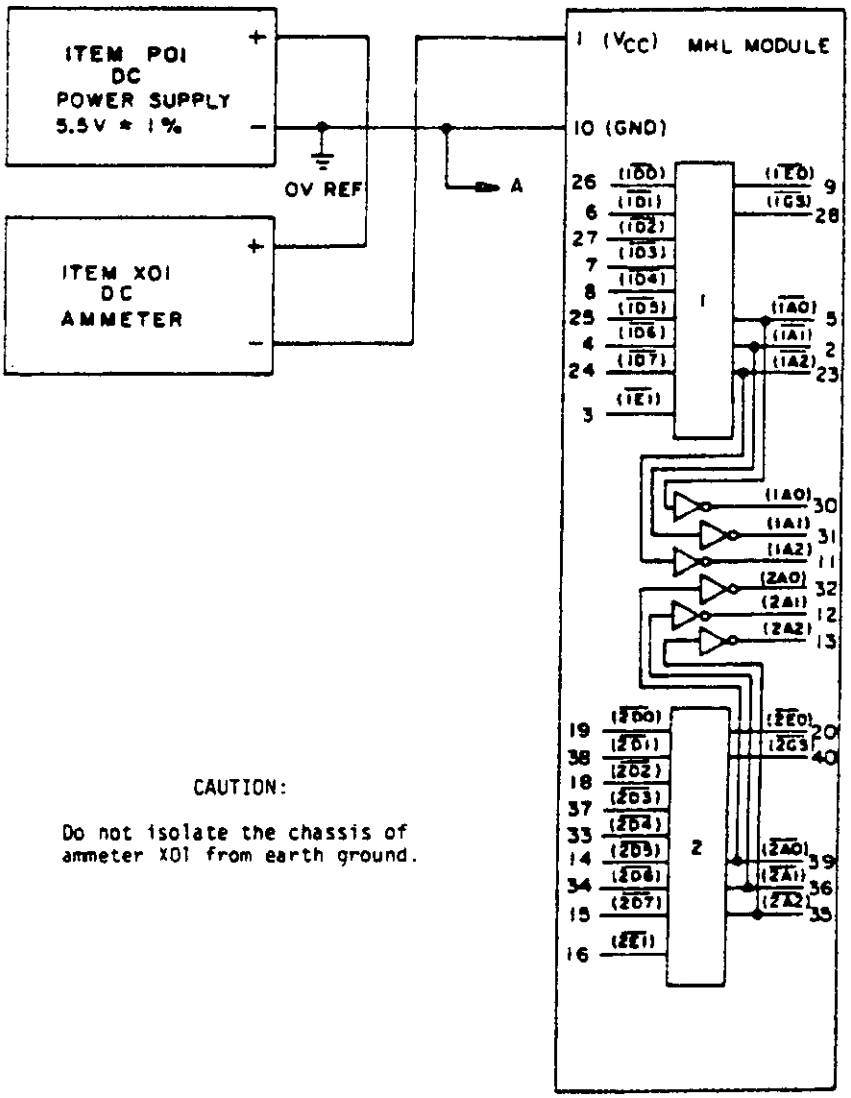



FIGURE 16. Power supply current (low-level outputs and high-level outputs) test circuit for table XIII.



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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| M1 | $\overline{IEO}$ Transition time ($t_{TLH}$), $C_L$ = 50 pF | 4,6,7, 8,25, 26,27 | 3 | 24 | 9 | Open | 4 | 30 | ns | 17,18 |
| M2 | $\overline{1GS}$ " | " | " | " | 28 | " | " | " | " | " |
| M3 | 1A0 " | " | " | " | 5 | " | " | " | " | " |
| M4 | 1A1 " | " | " | " | 2 | " | " | " | " | " |
| M5 | $\overline{1A2}$ " | " | " | " | 23 | " | " | " | " | " |
| M6 | 1A0 " | " | " | " | 30 | 5 | " | " | " | " |
| M7 | 1A1 " | " | " | " | 31 | 2 | " | " | " | " |
| M8 | 1A2 " | " | " | " | 11 | 23 | " | " | " | " |
| M9 | $\overline{2EO}$ " | 11,18, 19,33, 34,37, 38 | 16 | 15 | 20 | Open | " | " | " | " |
| M10 | $\overline{2GS}$ " | " | " | " | 40 | " | " | " | " | " |
| M11 | $\overline{2A0}$ " | " | " | " | 39 | " | " | " | " | " |
| M12 | $\overline{2A1}$ " | " | " | " | 36 | " | " | " | " | " |
| M13 | $\overline{2A2}$ " | " | " | " | 35 | " | " | " | " | " |
| M14 | 2A0 " | " | " | " | 32 | 39 | " | " | " | " |
| M15 | 2A1 " | " | " | " | 12 | 36 | " | " | " | " |
| M16 | 2A2 " | " | " | " | 13 | 35 | " | " | " | " |
| M17 | $\overline{IEO}$ Transition time ($t_{TLH}$), $C_L$ = 150 pF | 4,6,7, 8,25, 26,27 | 3 | 24 | 9 | Open | " | 50 | " | " |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| M18 | $\overline{1GS}$ | Transition time ($t_{TLH}$), $C_L$ = 150 pF | 4,6,7, 8,25, 26,27 | 3 | 24 | 28 | Open | 4 | 50 | ns | 17,18 |
| M19 | $\overline{1A0}$ | " | " | " | " | 5 | " | " | " | " | " |
| M20 | $\overline{1A1}$ | " | " | " | " | 2 | " | " | " | " | " |
| M21 | $\overline{1A2}$ | " | " | " | " | 23 | " | " | " | " | " |
| M22 | $\overline{1A0}$ | " | " | " | " | 30 | 5 | " | " | " | " |
| M23 | 1A1 | " | " | " | " | 31 | 2 | " | " | " | " |
| M24 | 1A2 | " | " | " | " | 11 | 23 | " | " | " | " |
| M25 | $\overline{2E0}$ | " | 14,18, 19,33, 34,37 38 | 16 | 15 | 20 | Open | " | " | " | " |
| M26 | $\overline{2GS}$ | " | " | " | " | 40 | " | " | " | " | " |
| M27 | $\overline{2A0}$ | " | " | " | " | 39 | " | " | " | " | " |
| M28 | $\overline{2A1}$ | " | " | " | " | 36 | " | " | " | " | " |
| M29 | $\overline{2A2}$ | " | " | " | " | 35 | " | " | " | " | " |
| M30 | 2A0 | " | " | " | " | 32 | 39 | " | " | " | " |
| M31 | 2A1 | " | " | " | " | 12 | 36 | " | " | " | " |
| M32 | 2A2 | " | " | " | " | 13 | 35 | " | " | " | " |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| N1 | $\overline{1E0}$ Transition time ($t_{THL}$), $C_L$ = 50 pF | 4,6,7, 8,25, 26,27 | 3 | 24 | 9 | Open | 4 | 13 | ns | 17,18 |
| N2 | $\overline{1G5}$ " | " | " | " | 28 | " | " | " | " | " |
| N3 | $\overline{1A0}$ " | " | " | " | 5 | " | " | " | " | " |
| N4 | $\overline{1A1}$ " | " | " | " | 2 | " | " | " | " | " |
| N5 | $\overline{1A2}$ " | " | " | " | 23 | " | " | " | " | " |
| N6 | 1A0 " | " | " | " | 30 | 5 | " | " | " | " |
| N7 | 1A1 " | " | " | " | 31 | 2 | " | " | " | " |
| N8 | 1A2 " | " | " | " | 11 | 23 | " | " | " | " |
| N9 | $\overline{2E0}$ " | 14,18, 19,33, 34,37, 38 | 16 | 15 | 20 | Open | " | " | " | " |
| N10 | $\overline{2G5}$ " | " | " | " | 40 | " | " | " | " | " |
| N11 | $\overline{2A0}$ " | " | " | " | 39 | " | " | " | " | " |
| N12 | $\overline{2A1}$ " | " | " | " | 36 | " | " | " | " | " |
| N13 | $\overline{2A2}$ " | " | " | " | 35 | " | " | " | " | " |
| N14 | 2A0 " | " | " | " | 32 | 39 | " | " | " | " |
| N15 | 2A1 " | " | " | " | 12 | 36 | " | " | " | " |
| N16 | 2A2 " | " | " | " | 13 | 35 | " | " | " | " |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| N17 | $\overline{1E0}$ Transition time ($t_{THL}$), $C_L$ = 150 pF | 4,6,7, 8,25, 26,27 | 3 | 24 | 9 | Open | 4 | 20 | ns | 17,18 |
| N18 | $\overline{1GS}$ " | " | " | " | 28 | " | " | " | " | " |
| N19 | $\overline{1A0}$ " | " | " | " | 5 | " | " | " | " | " |
| N20 | $\overline{1A1}$ " | " | " | " | 2 | " | " | " | " | " |
| N21 | $\overline{1A2}$ " | " | " | " | 23 | " | " | " | " | " |
| N22 | 1A0 " | " | " | " | 30 | 5 | " | " | " | " |
| N23 | 1A1 " | " | " | " | 31 | 2 | " | " | " | " |
| N24 | 1A2 " | " | " | " | 11 | 23 | " | " | " | " |
| N25 | $\overline{2E0}$ " | 14,18, 19,33, 34,37, 38 | 16 | 15 | 20 | Open | " | " | " | " |
| N26 | $\overline{2GS}$ " | " | " | " | 40 | " | " | " | " | " |
| N27 | $\overline{2A0}$ " | " | " | " | 39 | " | " | " | " | " |
| N28 | $\overline{2A1}$ " | " | " | " | 36 | " | " | " | " | " |
| N29 | $\overline{2A2}$ " | " | " | " | 35 | " | " | " | " | " |
| N30 | 2A0 " | " | " | " | 32 | 39 | " | " | " | " |
| N31 | 2A1 " | " | " | " | 12 | 36 | " | " | " | " |
| N32 | 2A2 " | " | " | " | 13 | 35 | " | " | " | " |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| P1 | $\overline{1D0}$ to $\overline{1E0}$ Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | 4,6,7, 8,24, 25,27 | 3 | 26 | 9 | Open | 2 | 12 | ns | 17,19 |
| P2 | $\overline{1D1}$ to $\overline{1E0}$ " | 4,7,8, 24,25, 26,27 | " | 6 | " | " | " | " | " | " |
| P3 | $\overline{1D2}$ to $\overline{1E0}$ " | 4,6,7, 8,24, 25,26 | " | 27 | " | " | " | " | " | " |
| P4 | $\overline{1D3}$ to $\overline{1E0}$ " | 4,6,8, 24,25, 26,27 | " | 7 | " | " | " | " | " | " |
| P5 | $\overline{2D0}$ to $\overline{2E0}$ " | 14,15, 18,33, 34,37, 38 | 16 | 19 | 20 | " | " | " | " | " |
| P6 | $\overline{2D1}$ to $\overline{2E0}$ " | 14,15, 18,19, 33,34, 37 | " | 38 | " | " | " | " | " | " |
| P7 | $\overline{2D2}$ to $\overline{2E0}$ " | 14,15, 19,33, 34,37, 38 | " | 18 | " | " | " | " | " | " |
| P8 | $\overline{2D3}$ to $\overline{2E0}$ " | 14,15, 18,19, 33,34, 38 | " | 37 | " | " | " | " | " | " |
| P9 | $\overline{1E1}$ to $\overline{1E0}$ " | 4,6,7, 8,24, 25,26, 27 | Open | 3 | 9 | " | 5 | 16 | " | 17,20 |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| P10 | $\overline{2ET}$ to $\overline{2E0}$ Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | 14,15, 18,19, 33,34, 37,38 | Open | 16 | 20 | Open | 5 | 18 | ns | 17,20 |
| P11 | $\overline{1D0}$ to $\overline{1GS}$ " | 4,6,7, 8,24, 25,27 | 3 | 26 | 28 | " | 8 | 44 | " | " |
| P12 | $\overline{1D1}$ to $\overline{1GS}$ " | 4,7,8, 24,25, 26,27 | " | 6 | " | " | " | " | " | " |
| P13 | $\overline{1D2}$ to $\overline{1GS}$ " | 4,6,7, 8,24, 25,26 | " | 27 | " | " | " | " | " | " |
| P14 | $\overline{1D3}$ to $\overline{1GS}$ " | 4,6,8, 24,25, 26,27 | " | 7 | " | " | " | " | " | " |
| P15 | $\overline{1D7}$ to $\overline{1GS}$ " | 4,6,7, 8,25, 26,27 | " | 24 | " | " | " | " | " | " |
| P16 | $\overline{2D0}$ to $\overline{2GS}$ " | 14,15, 18,33, 34,37, 38 | 16 | 19 | 40 | " | " | " | " | " |
| P17 | $\overline{2D1}$ to $\overline{2GS}$ " | 14,15, 18,19, 33,34, 37 | " | 38 | " | " | " | " | " | " |
| P18 | $\overline{2D2}$ to $\overline{2GS}$ " | 14,15, 19,33, 34,38 | " | 18 | " | " | " | " | " | " |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| P19 | $\overline{2D3}$ to $\overline{2GS}$ Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | 14,15, 18,19, 33,34, 38 | 16 | 37 | 40 | Open | 8 | 44 | ns | 17,20 |
| P20 | $\overline{2D7}$ to $\overline{2GS}$ " | 14,18, 19,33, 34,37, 38 | 16 | 15 | 40 | " | 8 | 44 | " | " |
| P21 | $\overline{1E1}$ to $\overline{1GS}$ " | 4,6,7, 8,25, 26,27 | 24 | 3 | 28 | " | 4 | 19 | " | " |
| P22 | $\overline{2E1}$ to $\overline{2GS}$ " | 14,18, 19,33, 34,37, 38 | 15 | 16 | 40 | " | 4 | 19 | " | " |
| P23 | $\overline{1D1}$ to $\overline{1A0}$ " | 4,7,8, 24,25, 27 | 3,26 | 6 | 5 | " | 8 | 33 | " | " |
| P24 | $\overline{1D2}$ to $\overline{1A0}$ " | 4,7,8, 24,25, 26 | 3,6 | 27 | " | " | " | " | " | 17,19 |
| P25 | $\overline{1D3}$ to $\overline{1A0}$ " | 4,6,8, 24,25, 26 | 3,27 | 7 | " | " | " | " | " | 17,20 |
| P26 | $\overline{1D4}$ to $\overline{1A0}$ " | 4,6,24, 25,26, 27 | 3,7 | 8 | " | " | " | " | " | 17,19 |
| P27 | $\overline{1D5}$ to $\overline{1A0}$ " | 4,6,7, 24,26, 27 | 3,8 | 25 | " | " | " | " | " | 17,20 |
| P28 | $\overline{1D6}$ to $\overline{1A0}$ " | 6,7,8, 24,26, 27 | 3,25 | 4 | " | " | " | " | " | 17,19 |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| P29 | $\overline{ID7}$ to $\overline{TA0}$ Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | 4,6,7, 8,25, 26,27 | 3 | 24 | 5 | Open | 8 | 33 | ns | 17,20 |
| P30 | $\overline{ID2}$ to $\overline{TA1}$ " | 4,7,8, 24,25, 26 | 3,6 | 27 | 2 | " | " | " | " | " |
| P31 | $\overline{ID3}$ to $\overline{TA1}$ " | 4,8,24, 25,26, 27 | 3,6 | 7 | " | " | " | " | " | " |
| P32 | $\overline{ID4}$ to $\overline{TA1}$ " | 4,6,24, 25,26, 27 | 3,7 | 8 | " | " | " | " | " | 17,19 |
| P33 | $\overline{ID5}$ to $\overline{TA1}$ " | 4,6,8, 24,26, 27 | 3,7 | 25 | " | " | " | " | " | 17,19 |
| P34 | $\overline{ID6}$ to $\overline{TA1}$ " | 6,7,8, 24,25, 26,27 | 3 | 4 | " | " | " | " | " | 17,20 |
| P35 | $\overline{ID7}$ to $\overline{TA1}$ " | 4,6,7, 8,25, 26,27 | 3 | 24 | " | " | " | " | " | " |
| P36 | $\overline{ID4}$ to $\overline{TA2}$ " | 4,6,24, 25,26, 27 | 3,7 | 8 | 23 | " | " | " | " | " |
| P37 | $\overline{ID5}$ to $\overline{TA2}$ " | 4,6,8, 24,26, 27 | 3,7 | 25 | 23 | " | " | " | " | " |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| P38 | $\overline{1D6}$ to $\overline{1A2}$ Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | 6,8,24, 25,26, 27 | 3,7 | 4 | 23 | Open | 8 | 33 | ns | 17,20 |
| P39 | $\overline{1D7}$ to $\overline{1A2}$ " | 4,6,7, 8,25, 26,27 | 3 | 24 | 23 | " | " | " | " | " |
| P40 | $\overline{2D1}$ to $\overline{2A0}$ " | 14,15, 18,33, 34,37 | 16,19 | 38 | 39 | " | " | " | " | " |
| P41 | $\overline{2D2}$ to $\overline{2A0}$ " | 14,15, 19,33, 34,37 | 16,38 | 18 | " | " | " | " | " | 17,19 |
| P42 | $\overline{2D3}$ to $\overline{2A0}$ " | 14,15, 19,33, 34,38 | 16,18 | 37 | " | " | " | " | " | 17,20 |
| P43 | $\overline{2D4}$ to $\overline{2A0}$ " | 14,15, 18,19, 34,38 | 16,37 | 33 | " | " | " | " | " | 17,19 |
| P44 | $\overline{2D5}$ to $\overline{2A0}$ " | 15,18, 19,34, 37,38 | 16,33 | 14 | " | " | " | " | " | 17,20 |
| P45 | $\overline{2D6}$ to $\overline{2A0}$ " | 15,18, 19,33, 37,38 | 14,16 | 34 | " | " | " | " | " | 17,19 |
| P46 | $\overline{2D7}$ to $\overline{2A0}$ " | 14,18, 19,33, 34,37, 38 | 16 | 15 | " | " | " | " | " | 17,20 |
| P47 | $\overline{2D2}$ to $\overline{2A1}$ " | 14,15, 19,33, 34,37 | 16,36 | 18 | 36 | " | " | " | " | 17,20 |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| P48 | $\overline{2D3}$ to $\overline{2A1}$ Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | 14,15, 18,19, 33,34 | 16,38 | 37 | 36 | Open | 8 | 33 | ns | 17,20 |
| P49 | $\overline{2D4}$ to $\overline{2A1}$ " | 14,15, 18,19, 34,38 | 16,37 | 33 | " | " | " | " | " | 17,19 |
| P50 | $\overline{2D5}$ to $\overline{2A1}$ " | 15,18, 19,33, 34,38 | 16,37 | 14 | " | " | " | " | " | 17,19 |
| P51 | $\overline{2D6}$ to $\overline{2A1}$ " | 14,15, 18,19, 33,37, 38 | 16 | 34 | " | " | " | " | " | 17,20 |
| P52 | $\overline{2D7}$ to $\overline{2A1}$ " | 14,18, 19,33, 34,37, 38 | 16 | 15 | " | " | " | " | " | " |
| P53 | $\overline{2D4}$ to $\overline{2A2}$ " | 14,15, 18,19, 34,38 | 16,37 | 33 | 35 | " | " | " | " | " |
| P54 | $\overline{2D5}$ to $\overline{2A2}$ " | 15,18, 19,33, 34,38 | 16,37 | 14 | " | " | " | " | " | " |
| P55 | $\overline{2D6}$ to $\overline{2A2}$ " | 14,15, 18,19, 33,38 | 16,37 | 34 | " | " | " | " | " | " |
| P56 | $\overline{2D7}$ to $\overline{2A2}$ " | 14,18, 19,33, 34,37 38 | 16 | 15 | " | " | " | " | " | " |
| P57 | $\overline{1E1}$ to $\overline{1A0}$ " | 4,6,7, 8,25, 26,27 | 24 | 3 | 5 | " | 7 | 21 | " | " |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial values 25°C 45°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| P58 | $\overline{1ET}$ to $\overline{1AT}$ Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | 4,6,7, 8,25, 26,27 | 24 | 3 | 2 | Open | 7 | 21 | ns | 17,20 |
| P59 | $\overline{1ET}$ to $\overline{1AZ}$ " | 4,6,7, 8,25, 26,27 | 24 | 3 | 23 | " | " | " | " | " |
| P60 | $\overline{2ET}$ to $\overline{2AO}$ " | 14,18, 19,33, 34,37, 38 | 15 | 16 | 39 | " | " | " | " | " |
| P61 | $\overline{2ET}$ to $\overline{2AI}$ " | 14,18, 19,33, 34,37, 38 | 15 | 16 | 36 | " | " | " | " | " |
| P62 | $\overline{2ET}$ to $\overline{2A2}$ " | 14,18, 19,33, 34,37, 38 | 15 | 16 | 35 | " | " | " | " | " |
| P63 | $\overline{1D1}$ to $\overline{1AO}$ " | 4,7,8, 24,25, 27 | 3,26 | 6 | 30 | 5 | 13 | 54 | " | 17,19 |
| P64 | $\overline{1D7}$ to 1AO " | 4,6,7, 8,25, 26,27 | 3 | 24 | 30 | 5 | " | " | " | " |
| P65 | $\overline{1D2}$ to 1A1 " | 4,7,8, 24,25, 26 | 3,6 | 27 | 31 | 2 | " | " | " | " |
| P66 | $\overline{1D6}$ to 1A1 " | 6,7,8, 24,25, 26,27 | 3 | 4 | 31 | 2 | " | " | " | " |
| P67 | $\overline{1D4}$ to 1A2 " | 4,6,24, 25,26, 27 | 3,7 | 8 | 11 | 23 | " | " | " | " |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| P68 | $\overline{1D5}$ to 1A2 Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | 4,6,8, 24,26, 27 | 3,7 | 25 | 11 | 23 | 13 | 54 | ns | 17,19 |
| P69 | $\overline{2D1}$ to 2A0 " | 14,15, 18,33, 34,37, | 16,19 | 38 | 32 | 39 | " | " | " | " |
| P70 | $\overline{2D7}$ to 2A0 " | 14,18, 19,33, 34,37, 38 | 16 | 15 | 32 | 39 | " | " | " | " |
| P71 | $\overline{2D2}$ to 2A1 " | 14,15, 19,33, 34,37 | 16,38 | 18 | 12 | 36 | " | " | " | " |
| P72 | $\overline{2D6}$ to 2A1 " | 14,15, 18,19, 33,37, 38 | 16 | 34 | 12 | 36 | " | " | " | " |
| P73 | $\overline{2D4}$ to 2A2 " | 14,15, 18,19, 34,38 | 16,37 | 33 | 13 | 35 | " | " | " | " |
| P74 | $\overline{2D5}$ to 2A2 " | 15,18, 19,33, 34,38 | 16,37 | 14 | 13 | 35 | " | " | " | " |
| P75 | $\overline{1E1}$ to 1A0 " | 4,6,7, 8,25, 26,27 | 24 | 3 | 30 | 5 | 10 | 51 | " | " |
| P76 | $\overline{1E1}$ to 1A1 " | 4,6,7, 8,25, 26,27 | 24 | 3 | 31 | 2 | " | " | " | " |
| P77 | $\overline{1E1}$ to 1A2 " | 4,6,7, 8,25, 26,27 | 24 | 3 | 11 | 23 | " | " | " | " |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| P78 | $\overline{2ET}$ to 2A0 Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | 14,18, 19,33, 34,37, 38 | 15 | 16 | 32 | 39 | 10 | 51 | ns | 17,19 |
| P79 | $\overline{2ET}$ to 2A1 " | 14,18, 19,33, 34,37, 38 | 15 | 16 | 12 | 36 | " | " | " | " |
| P80 | $\overline{2ET}$ to 2A2 " | 14,18, 19,33, 34,37, 38 | 15 | 16 | 13 | 35 | " | " | " | " |
| P81 | $\overline{1D0}$ to $\overline{1E0}$ Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | 4,6,7, 8,24, 25,27, | 3 | 26 | 9 | Open | 4 | 17 | " | " |
| P82 | $\overline{1D1}$ to $\overline{1E0}$ " | 4,7,8, 24,25, 26,27, | " | 6 | " | " | " | " | " | " |
| P83 | $\overline{1D2}$ to $\overline{1E0}$ " | 4,6,7, 8,24, 25,26, | " | 27 | " | " | " | " | " | " |
| P84 | $\overline{1D3}$ to $\overline{1E0}$ " | 4,6,8, 24,25, 26,27, | " | 7 | " | " | " | " | " | " |
| P85 | $\overline{2D0}$ to $\overline{2E0}$ " | 14,15, 18,33, 34,37, 38 | 16 | 19 | 20 | " | " | " | " | " |
| P86 | $\overline{2D1}$ to $\overline{2E0}$ " | 14,15, 18,19, 33,34, 37 | 16 | 38 | 20 | " | " | " | " | " |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| P87 | $\overline{2D2}$ to $\overline{2E0}$ Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | 14,15, 19,33, 34,37, 38 | 16 | 18 | 20 | Open | 4 | 17 | ns | 17,19 |
| P88 | $\overline{2D3}$ to $\overline{2E0}$ " | 14,15, 18,19, 33,34, 38 | 16 | 37 | 20 | " | 4 | 17 | " | 17,19 |
| P89 | $\overline{1E1}$ to $\overline{1E0}$ " | 4,6,7, 8,24, 25,26, 27 | Open | 3 | 9 | " | 5 | 24 | " | 17,20 |
| P90 | $\overline{2E1}$ to $\overline{2E0}$ " | 14,15, 18,19, 33,34, 37,38 | Open | 16 | 20 | " | 5 | 24 | " | " |
| P91 | $\overline{1D0}$ to $\overline{1G5}$ " | 4,6,7, 8,24, 25,27 | 3 | 26 | 28 | " | 8 | 50 | " | " |
| P92 | $\overline{1D1}$ to $\overline{1G5}$ " | 4,7,8, 24,25, 26,27 | " | 6 | " | " | " | " | " | " |
| P93 | $\overline{1D2}$ to $\overline{1G5}$ " | 4,6,7, 8,24, 25,26 | " | 27 | " | " | " | " | " | " |
| P94 | $\overline{1D3}$ to $\overline{1G5}$ " | 4,6,8, 24,25, 26,27 | " | 7 | " | " | " | " | " | " |
| P95 | $\overline{1D7}$ to $\overline{1G5}$ " | 4,6,7, 8,25, 26,27 | " | 24 | " | " | " | " | " | " |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| P96 | $\overline{2D0}$ to $\overline{2G5}$ Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | 14,15, 18,33, 34,37, 38 | 16 | 19 | 40 | Open | 8 | 50 | ns | 17,20 |
| P97 | $\overline{2D1}$ to $\overline{2G5}$ " | 14,15, 18,19, 33,34, 38 | " | 38 | " | " | " | " | " | " |
| P98 | $\overline{2D2}$ to $\overline{2G5}$ " | 14,15, 19,33, 34,37, 38 | " | 18 | " | " | " | " | " | " |
| P99 | $\overline{2D3}$ to $\overline{2G5}$ " | 14,15, 18,19, 33,34, 38 | " | 37 | " | " | " | " | " | " |
| P100 | $\overline{2D7}$ to $\overline{2G5}$ " | 14,18, 19,33, 34,37, 38 | " | 15 | " | " | " | " | " | " |
| P101 | $\overline{1E1}$ to $\overline{1G5}$ " | 4,6,7, 18,25, 26,27 | 24 | 3 | 28 | " | 4 | 25 | " | " |
| P102 | $\overline{2E1}$ to $\overline{2G5}$ " | 14,18, 19,33, 34,37, 38 | 15 | 16 | 40 | " | 4 | 25 | " | " |
| P103 | $\overline{1D1}$ to $\overline{1A0}$ " | 4,7,8, 24,25, 27 | 3,26 | 6 | 5 | " | 10 | 39 | " | " |
| P104 | $\overline{1D2}$ to $\overline{1A0}$ " | 4,7,8, 24,25, 26 | 3,6 | 27 | 5 | " | 10 | 39 | " | 17,19 |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| P105 | $\overline{ID3}$ to $\overline{IA0}$ Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | 4,6,8, 24,25, 26 | 3,27 | 7 | 5 | Open | 10 | 39 | ns | 17,20 |
| P106 | $\overline{ID4}$ to $\overline{IA0}$ " | 4,6, 24,25, 26,27 | 3,7 | 8 | " | " | " | " | " | 17,19 |
| P107 | $\overline{ID5}$ to $\overline{IA0}$ " | 4,6,7, 24,26, 27 | 3,8 | 25 | " | " | " | " | " | 17,20 |
| P108 | $\overline{ID6}$ to $\overline{IA0}$ " | 6,7,8, 24,26, 27 | 3,25 | 4 | " | " | " | " | " | 17,19 |
| P109 | $\overline{ID7}$ to $\overline{IA0}$ " | 4,6,7, 8,25, 26,27 | 3 | 24 | " | " | " | " | " | 17,20 |
| P110 | $\overline{ID2}$ to $\overline{IA1}$ " | 4,7,8, 24,25, 26 | 3,6 | 27 | 2 | " | " | " | " | 17,20 |
| P111 | $\overline{ID3}$ to $\overline{IA1}$ " | 4,8,24, 25,26, 27 | 3,6 | 7 | " | " | " | " | " | 17,20 |
| P112 | $\overline{ID4}$ to $\overline{IA1}$ " | 4,6,24, 25,26, 27 | 3,7 | 8 | " | " | " | " | " | 17,19 |
| P113 | $\overline{ID5}$ to $\overline{IA1}$ " | 4,6,8, 24,26, 27 | 3,7 | 25 | " | " | " | " | " | 17,19 |
| P114 | $\overline{ID6}$ to $\overline{IA1}$ " | 6,7,8, 24,25, 26,27 | 3 | 4 | " | " | " | " | " | 17,20 |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| P115 | 107 to 2A1 Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | 1,5,7, 8,23, 25,27 | 3 | 24 | 2 | 0 ten | 10 | 39 | ns | 17,20 |
| P116 | 104 to 2A2 " | 1,5,24, 25,26, 27 | 3,7 | 6 | 23 | " | " | " | " | " |
| P117 | 105 to 2A2 " | 1,5,7, 24,26, 27 | 3,7 | 25 | " | " | " | " | " | " |
| P119 | 106 to 2A2 " | 6,8,24, 25,25, 27 | 3,7 | 4 | " | " | " | " | " | " |
| P119 | 107 to 2A2 " | 4,6,7, 13,25, 26,27 | 3 | 24 | " | " | " | " | " | " |
| P120 | 201 to 2A0 " | 11,15, 18,33, 34,37 | 16,19 | 38 | 39 | " | " | " | " | " |
| P121 | 202 to 2A0 " | 14,15, 19,33, 34,37 | 16,38 | 18 | " | " | " | " | " | 17,19 |
| P122 | 203 to 2A0 " | 14,15, 19,33, 34,38 | 16,18 | 37 | " | " | " | " | " | 17,20 |
| P123 | 204 to 2A0 " | 14,15, 18,19, 34,38 | 16,37 | 33 | " | " | " | " | " | 17,19 |
| P124 | 205 to 2A0 " | 15,18, 19,34, 37,38 | 16,33 | 14 | " | " | " | " | " | 17,20 |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| P125 | $\overline{2D6}$ to $\overline{2A0}$ Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | 15,18, 19,33, 37,38 | 14,16 | 34 | 39 | Open | 10 | 39 | ns | 17,19 |
| P126 | $\overline{2D7}$ to $\overline{2A0}$ " | 14,18, 19,33, 34,37, 38 | 16 | 15 | 39 | Open | " | " | " | 17,20 |
| P127 | $\overline{2D2}$ to $\overline{2A1}$ " | 14,15, 19,33, 34,37 | 16,38 | 18 | 36 | " | " | " | " | 17,20 |
| P128 | $\overline{2D3}$ to $\overline{2A1}$ " | 14,15, 18,19, 33,34 | 16,38 | 37 | " | " | " | " | " | 17,20 |
| P129 | $\overline{2D4}$ to $\overline{2A1}$ " | 14,15, 18,19, 34,38 | 16,37 | 33 | " | " | " | " | " | 17,19 |
| P130 | $\overline{2D5}$ to $\overline{2A1}$ " | 15,18, 19,33, 34,38 | 16,37 | 14 | " | " | " | " | " | 17,19 |
| P131 | $\overline{2D6}$ to $\overline{2A1}$ " | 14,15, 18,19, 33,37, 38 | 16 | 34 | " | " | " | " | " | 17,20 |
| P132 | $\overline{2D7}$ to $\overline{2A1}$ " | 14,18, 19,33, 34,37, 38 | 16 | 15 | " | " | " | " | " | 17,20 |
| P133 | $\overline{2D4}$ to $\overline{2A2}$ " | 14,15, 18,19, 34,38 | 16,37 | 33 | 35 | " | " | " | " | 17,20 |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| P134 | $\overline{2D5}$ to $\overline{2A2}$ Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | 15,18, 19,33, 34,38 | 16,37 | 14 | 35 | Open | 10 | 39 | ns | 17,20 |
| P135 | $\overline{2D6}$ to $\overline{2A2}$ " | 14,15, 18,19, 33,38 | 16,37 | 34 | " | " | " | " | " | " |
| P136 | $\overline{2D7}$ to $\overline{2A2}$ " | 14,18, 19,33, 34,37, 38 | 16 | 15 | " | " | " | " | " | " |
| P137 | $\overline{1E1}$ to $\overline{1A0}$ " | 4,6,7, 8,25, 26,27 | 24 | 3 | 5 | " | 7 | 27 | " | " |
| P138 | $\overline{1E1}$ to $\overline{1A1}$ " | 4,6,7, 8,25, 26,27 | 24 | 3 | 2 | " | " | " | " | " |
| P139 | $\overline{1E1}$ to $\overline{1A2}$ " | 4,6,7, 8,25, 26,27 | 24 | 3 | 23 | " | " | " | " | " |
| P140 | $\overline{2E1}$ to $\overline{2A0}$ " | 14,18, 19,33, 34,37, 38 | 15 | 16 | 39 | " | " | " | " | " |
| P141 | $\overline{2E1}$ to $\overline{2A1}$ " | 14,18, 19,33, 34,37, 38 | " | " | 36 | " | " | " | " | " |
| P142 | $\overline{2E1}$ to $\overline{2A2}$ " | 14,18, 19,33, 34,37, 38 | " | " | 35 | " | " | " | " | " |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| P143 | $\overline{1D1}$ to 1A0 Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | 4,7,8, 24,25, 27 | 3,26 | 6 | 30 | 5 | 13 | 66 | ns | 17,19 |
| P144 | $\overline{1D7}$ to 1A0 " | 4,6,7, 8,25, 26,27 | 3 | 24 | 30 | 5 | " | " | " | " |
| P145 | $\overline{1D2}$ to 1A1 " | 4,7,8, 24,25, 26 | 3,6 | 27 | 31 | 2 | " | " | " | " |
| P146 | $\overline{1D6}$ to 1A1 " | 6,7,8, 24,25, 26,27 | 3 | 4 | 31 | 2 | " | " | " | " |
| P147 | $\overline{1D4}$ to 1A2 " | 4,6,24, 25,26, 27 | 3,7 | 8 | 11 | 23 | " | " | " | " |
| P148 | $\overline{1D5}$ to 1A2 " | 4,6,8, 24,26, 27 | 3,7 | 25 | 11 | 23 | " | " | " | " |
| P149 | $\overline{2D1}$ to 2A0 " | 14,15, 18,33, 34,37 | 16,19 | 38 | 32 | 39 | " | " | " | " |
| P150 | $\overline{2D7}$ to 2A0 " | 14,18, 19,33, 34,37, 38 | 16 | 15 | 32 | 39 | " | " | " | " |
| P151 | $\overline{2D2}$ to 2A1 " | 14,15, 19,33, 34,37 | 16,38 | 18 | 12 | 36 | " | " | " | " |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| P152 | $\overline{2D6}$ to 2A1 Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | 14,15, 18,19, 33,37, 38 | 16 | 34 | 12 | 36 | 13 | 66 | ns | 17,19 |
| P153 | $\overline{2D4}$ to 2A2 " | 14,15, 18,19, 34,38 | 16,37 | 33 | 13 | 35 | 13 | 66 | " | " |
| P154 | $\overline{2D5}$ to 2A2 " | 15,18, 19,33, 34,38 | 16,37 | 14 | 13 | 35 | 13 | 66 | " | " |
| P155 | $\overline{1ET}$ to 1A0 " | 4,6,7, 8,25, 26,27 | 24 | 3 | 30 | 5 | 10 | 63 | " | " |
| P156 | $\overline{1ET}$ to 1A1 " | 4,6,7, 8,25, 26,27 | 24 | 3 | 31 | 2 | " | " | " | " |
| P157 | $\overline{1ET}$ to 1A2 " | 4,6,7, 8,25, 26,27 | 24 | 3 | 11 | 23 | " | " | " | " |
| P158 | $\overline{2ET}$ to 2A0 " | 14,18, 19,33, 34,37, 38 | 15 | 16 | 32 | 39 | " | " | " | " |
| P159 | $\overline{2ET}$ to 2A1 " | 14,18, 19,33, 34,37, 38 | 15 | 16 | 12 | 36 | " | " | " | " |
| P160 | $\overline{2ET}$ to 2A2 " | 14,18, 19,33, 34,37, 38 | 15 | 16 | 13 | 35 | " | " | " | " |



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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| Q1 | $\overline{1D0}$ to $\overline{1E0}$ Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | 4,6,7, 8,24, 25,27 | 3 | 26 | 9 | Open | 4 | 28 | ns | 17,19 |
| Q2 | $\overline{1D4}$ to $\overline{1E0}$ " | 4,6,7, 24,25, 26,27 | 3 | 8 | " | " | " | " | " | " |
| Q3 | $\overline{1D5}$ to $\overline{1E0}$ " | 4,6,7, 8,24, 26,27 | " | 25 | " | " | " | " | " | " |
| Q4 | $\overline{1D6}$ to $\overline{1E0}$ " | 6,7,8, 24,25, 26,27 | " | 4 | " | " | " | " | " | " |
| Q5 | $\overline{1D7}$ to $\overline{1E0}$ " | 4,6,7, 8,25, 26,27 | " | 24 | " | " | " | " | " | " |
| Q6 | $\overline{2D0}$ to $\overline{2E0}$ " | 14,15, 18,33, 34,37, 38 | 16 | 19 | 20 | " | " | " | " | " |
| Q7 | $\overline{2D4}$ to $\overline{2E0}$ " | 14,15, 18,19, 34,37, 38 | " | 33 | " | " | " | " | " | " |
| Q8 | $\overline{2D5}$ to $\overline{2E0}$ " | 15,18, 19,33, 34,37, 38 | " | 14 | " | " | " | " | " | " |
| Q9 | $\overline{2D6}$ to $\overline{2E0}$ " | 14,15, 18,19, 33,37, 38 | " | 34 | " | " | " | " | " | " |
| Q10 | $\overline{2D7}$ to $\overline{2E0}$ " | 14,18, 19,33, 34,37, 38 | " | 15 | " | " | " | " | " | " |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| Q11 | $\overline{1E1}$ to $\overline{1E0}$ Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | 4,6,7, 8,24, 25,26, 27 | Open | 3 | 9 | Open | 8 | 39 | ns | 17,20 |
| Q12 | $\overline{2E1}$ to $\overline{2E0}$ " | 14,15, 18,19, 33,34, 37,38 | Open | 16 | 20 | " | " | 39 | " | " |
| Q13 | $\overline{1D0}$ to $\overline{1G5}$ " | 4,6,7, 8,24, 25,27, | 3 | 26 | 28 | " | " | 26 | " | " |
| Q14 | $\overline{1D4}$ to $\overline{1G5}$ " | 4,6,7, 24,25, 26,27, | " | 8 | " | " | " | " | " | " |
| Q15 | $\overline{1D5}$ to $\overline{1G5}$ " | 4,6,7, 8,24, 26,27 | " | 25 | " | " | " | " | " | " |
| Q16 | $\overline{1D6}$ to $\overline{1G5}$ " | 6,7,8, 24,25, 26,27 | " | 4 | " | " | " | " | " | " |
| Q17 | $\overline{1D7}$ to $\overline{1G5}$ " | 4,6,7, 8,25, 26,27, | " | 24 | " | " | " | " | " | " |
| Q18 | $\overline{2D0}$ to $\overline{2G5}$ " | 14,15, 18,33, 34,37, 38 | 16 | 19 | 40 | " | " | " | " | " |
| Q19 | $\overline{2D4}$ to $\overline{2G5}$ " | 14,15, 18,19, 34,37, 38 | " | 33 | " | " | " | " | " | " |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| Q20 | $\overline{2D5}$ to $\overline{2GS}$ Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | 15,18, 19,33, 34,37, 38 | 16 | 14 | 40 | Open | 8 | 26 | ns | 17,20 |
| Q21 | $\overline{2D6}$ to $\overline{2GS}$ " | 14,15, 18,19, 33,37, 38 | " | 34 | " | " | " | " | " | " |
| Q22 | $\overline{2D7}$ to $\overline{2GS}$ " | 14,18, 19,33, 34,37, 38 | " | 15 | " | " | " | " | " | " |
| Q23 | $\overline{1E1}$ to $\overline{1GS}$ " | 4,6,7, 8,25, 26,27 | 24 | 3 | 28 | " | 7 | 27 | " | " |
| Q24 | $\overline{2E1}$ to $\overline{2GS}$ " | 14,18, 19,33, 34,37, 38 | 15 | 16 | 40 | " | 7 | 27 | " | " |
| Q25 | $\overline{1D1}$ to $\overline{1A0}$ " | 4,7,8, 24,25, 27 | 3,26 | 6 | 5 | " | 8 | 30 | " | " |
| Q26 | $\overline{1D2}$ to $\overline{1A0}$ " | 4,7,8, 24,25, 26 | 3,6 | 27 | " | " | " | " | " | 17,19 |
| Q27 | $\overline{1D3}$ to $\overline{1A0}$ " | 4,6,8, 24,25, 26 | 3,27 | 7 | " | " | " | " | " | 17,20 |
| Q28 | $\overline{1D4}$ to $\overline{1A0}$ " | 4,6,24, 25,26, 27 | 3,7 | 8 | " | " | " | " | " | 17,19 |
| Q29 | $\overline{1D5}$ to $\overline{1A0}$ " | 4,6,7, 24,26, 27 | 3,8 | 25 | " | " | " | " | " | 17,20 |





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TABLE KIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| Q30 | $\overline{TD6}$ to $\overline{TA0}$ Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | 6,7,8, 24,26, 27 | 3,25 | 4 | 5 | Open | 8 | 30 | ns | 17,19 |
| Q31 | $\overline{TD7}$ to $\overline{TA0}$ " | 4,6,7, 8,25, 26,27 | 3 | 24 | " | " | " | " | " | 17,20 |
| Q32 | $\overline{TD2}$ to $\overline{TA1}$ " | 4,7,8, 24,25, 26 | 3,6 | 27 | 2 | " | " | " | " | 17,20 |
| Q33 | $\overline{TD3}$ to $\overline{TA1}$ " | 4,8,24, 25,26, 27 | 3,6 | 7 | 2 | " | " | " | " | 17,20 |
| Q34 | $\overline{TD4}$ to $\overline{TA1}$ " | 4,6,24, 25,26, 27 | 3,7 | 8 | " | " | " | " | " | 17,19 |
| Q35 | $\overline{TD5}$ to $\overline{TA1}$ " | 4,6,8, 24,26, 27 | 3,7 | 25 | " | " | " | " | " | 17,19 |
| Q36 | $\overline{TD6}$ to $\overline{TA1}$ " | 6,7,8, 24,25, 26,27 | 3 | 4 | " | " | " | " | " | 17,20 |
| Q37 | $\overline{TD7}$ to $\overline{TA1}$ " | 4,6,7, 8,25, 26,27 | 3 | 24 | " | " | " | " | " | " |
| Q38 | $\overline{TD4}$ to $\overline{TA2}$ " | 4,6,24, 25,26, 27 | 3,7 | 8 | 23 | " | " | " | " | " |
| Q39 | $\overline{TD5}$ to $\overline{TA2}$ " | 4,6,8, 24,26, 27 | 3,7 | 25 | " | " | " | " | " | " |
| Q40 | $\overline{TD6}$ to $\overline{TA2}$ " | 6,8,24, 25,26, 27 | 3,7 | 4 | " | " | " | " | " | " |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits -25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| Q41 | $\overline{1D7}$ to $\overline{1A2}$ Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | 4,6,7, 8,25, 26,27 | 3 | 24 | 23 | Open | 8 | 30 | ns | 17,20 |
| Q42 | $\overline{2D1}$ to $\overline{2A0}$ " | 14,15, 18,33, 34,37 | 16,19 | 38 | 39 | " | " | " | " | 17,20 |
| Q43 | $\overline{2D2}$ to $\overline{2A0}$ " | 14,15, 19,33, 34,37 | 16,38 | 18 | " | " | " | " | " | 17,19 |
| Q44 | $\overline{2D3}$ to $\overline{2A0}$ " | 14,15, 19,33, 34,38 | 16,18 | 37 | " | " | " | " | " | 17,20 |
| Q45 | $\overline{2D4}$ to $\overline{2A0}$ " | 14,15, 18,19, 34,38 | 16,37 | 33 | " | " | " | " | " | 17,19 |
| Q46 | $\overline{2D5}$ to $\overline{2A0}$ " | 15,18, 19,34, 37,38 | 16,33 | 14 | " | " | " | " | " | 17,20 |
| Q47 | $\overline{2D6}$ to $\overline{2A0}$ " | 15,18, 19,33, 37,38 | 14,16 | 34 | " | " | " | " | " | 17,19 |
| Q48 | $\overline{2D7}$ to $\overline{2A0}$ " | 14,18, 19,33, 34,37 38 | 16 | 15 | " | " | " | " | " | 17,20 |
| Q49 | $\overline{2D2}$ to $\overline{2A1}$ " | 14,15, 19,33, 34,37 | 16,38 | 18 | 36 | " | " | " | " | 17,20 |
| Q50 | $\overline{2D3}$ to $\overline{2A1}$ " | 14,15, 18,19, 33,34 | 16,38 | 37 | " | " | " | " | " | 17,20 |
| Q51 | $\overline{2D4}$ to $\overline{2A1}$ " | 14,15, 18,19, 34,38 | 16,37 | 33 | " | " | " | " | " | 17,19 |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| Q52 | $\overline{2D5}$ to $\overline{2A1}$ Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | 15,18, 19,33, 34,38 | 16,37 | 14 | 36 | Open | 8 | 30 | ns | 17,19 |
| Q53 | $\overline{2D6}$ to $\overline{2A1}$ " | 14,15, 18,19, 33,37, 38 | 16 | 34 | " | " | " | " | " | 17,20 |
| Q54 | $\overline{2D7}$ to $\overline{2A1}$ " | 14,18, 19,33, 34,37, 38 | 16 | 15 | " | " | " | " | " | " |
| Q55 | $\overline{2D4}$ to $\overline{2A2}$ " | 14,15, 18,19, 34,38 | 16,37 | 33 | 35 | " | " | " | " | " |
| Q56 | $\overline{2D5}$ to $\overline{2A2}$ " | 15,18, 19,33, 34,38 | " | 14 | " | " | " | " | " | " |
| Q57 | $\overline{2D6}$ to $\overline{2A2}$ " | 14,15, 18,19, 33,38 | " | 34 | " | " | " | " | " | " |
| Q58 | $\overline{2D7}$ to $\overline{2A2}$ " | 14,18, 19,33, 34,37, 38 | 16 | 15 | " | " | " | " | " | " |
| Q59 | $\overline{1E1}$ to $\overline{1A0}$ " | 4,6,7, 8,25, 26,27 | 24 | 3 | 5 | " | 7 | 27 | " | " |
| Q60 | $\overline{1E1}$ to $\overline{1A1}$ " | 4,6,7, 8,25, 26,27 | 24 | 3 | 2 | " | " | " | " | " |
| Q61 | $\overline{1E1}$ to $\overline{1A2}$ " | 4,6,7, 8,25, 26,27 | 24 | 3 | 23 | " | " | " | " | " |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| Q62 | $\overline{2E1}$ to $\overline{2A0}$ Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | 14,18, 19,33, 34,37, 38 | 15 | 16 | 39 | Open | 7 | 27 | ns | 17,20 |
| Q63 | $\overline{2E1}$ to $\overline{2A1}$ " | 14,18, 19,33, 34,37, 38 | 15 | 16 | 36 | " | " | " | " | " |
| Q64 | $\overline{2E1}$ to $\overline{2A2}$ " | 14,18, 19,33, 34,37, 38 | 15 | 16 | 35 | " | " | " | " | " |
| Q65 | $\overline{1D1}$ to $\overline{1A0}$ " | 4,7,8, 24,25, 27 | 3,26 | 6 | 30 | 5 | 13 | 54 | " | 17,19 |
| Q66 | $\overline{1D7}$ to 1A0 " | 4,6,7, 8,25, 26,27 | 3 | 24 | 30 | 5 | " | " | " | " |
| Q67 | $\overline{1D2}$ to 1A1 " | 4,7,8, 24,25, 26 | 3,6 | 27 | 31 | 2 | " | " | " | " |
| Q68 | $\overline{1D6}$ to 1A1 " | 6,7,8, 24,25, 26,27 | 3 | 4 | 31 | 2 | " | " | " | " |
| Q69 | $\overline{1D4}$ to 1A2 " | 4,6,24, 25,26, 27 | 3,7 | 8 | 11 | 23 | " | " | " | " |
| Q70 | $\overline{1D5}$ to 1A2 | 4,6,8, 24,26, 27 | 3,7 | 25 | 11 | 23 | " | " | " | " |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| Q71 | $\overline{2D1}$ to 2A0 Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | 14,15, 18,33, 34,37 | 16,19 | 38 | 32 | 39 | 13 | 54 | ns | 17,19 |
| Q72 | $\overline{2D7}$ to 2A0 " | 14,18, 19,33, 34,37, 38 | 16 | 15 | 32 | 39 | " | " | " | " |
| Q73 | $\overline{2D2}$ to 2A1 " | 14,15, 19,33, 34,37 | 16,38 | 18 | 12 | 36 | " | " | " | " |
| Q74 | $\overline{2D6}$ to 2A1 " | 14,15, 18,19, 33,37, 38 | 16 | 34 | 12 | 36 | " | " | " | " |
| Q75 | $\overline{2D4}$ to 2A2 " | 14,15, 18,19, 34,38 | 16,37 | 33 | 13 | 35 | " | " | " | " |
| Q76 | $\overline{2D5}$ to 2A2 " | 15,18, 19,33, 34,38 | 16,37 | 14 | 13 | 35 | " | " | " | " |
| Q77 | $\overline{1ET}$ to 1A0 " | 4,6,7, 8,25, 26,27 | 24 | 3 | 30 | 5 | 10 | 42 | " | " |
| Q78 | $\overline{1ET}$ to 1A1 " | 4,6,7, 8,25, 26,27 | 24 | 3 | 31 | 2 | " | " | " | " |
| Q79 | $\overline{1ET}$ to 1A2 " | 4,6,7, 8,25, 26,27 | 24 | 3 | 11 | 23 | " | " | " | " |
| Q80 | $\overline{2ET}$ to 2A0 " | 14,18, 19,33, 34,37, 38 | 15 | 16 | 32 | 39 | " | " | " | " |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C 45°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| Q81 | $\overline{2E1}$ to 2A1 Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | 14,18, 19,33, 34,37, 38 | 15 | 16 | 17 | 36 | 10 | 42 | ns | 17,19 |
| Q82 | $\overline{2E1}$ to 2A2 Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | 14,18, 19,33, 34,37, 38 | 15 | 16 | 13 | 35 | 10 | 42 | " | " |
| Q83 | $\overline{1D0}$ to $\overline{1E0}$ Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | 4,6,7, 8,24, 25,27 | 3 | 26 | 9 | Open | 4 | 33 | " | " |
| Q84 | $\overline{1D4}$ to $\overline{1E0}$ " | 4,6,7, 24,25, 26,27 | " | 8 | " | " | " | " | " | " |
| Q85 | $\overline{1D5}$ to $\overline{1E0}$ " | 4,6,7, 8,24, 26,27 | " | 25 | " | " | " | " | " | " |
| Q86 | $\overline{1D6}$ to $\overline{1E0}$ " | 6,7,8, 24,25, 26,27 | " | 4 | " | " | " | " | " | " |
| Q87 | $\overline{1D7}$ to $\overline{1E0}$ " | 4,6,7, 8,25, 26,27 | " | 24 | " | " | " | " | " | " |
| Q88 | $\overline{2D0}$ to $\overline{2E0}$ " | 14,15, 18,33, 34,37, 38 | 16 | 19 | 20 | " | " | " | " | " |
| Q89 | $\overline{2D4}$ to $\overline{2E0}$ " | 14,15, 18,19, 34,37, 38 | " | 11 | " | " | " | " | " | " |
| Q90 | $\overline{2D5}$ to $\overline{2E0}$ " | 15,18, 19,33, 34,37, | " | 14 | " | " | " | " | " | " |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| Q91 | $\overline{206}$ to $\overline{2E0}$ Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | 14,15, 18,19, 33,37, 38 | 16 | 34 | 20 | Open | 4 | 33 | ns | 17,19 |
| Q92 | $\overline{207}$ to $\overline{2E0}$ " | 14,18, 19,33, 34,37, 38 | 16 | 15 | 20 | " | 4 | 33 | " | 17,19 |
| Q93 | $\overline{1E1}$ to $\overline{1E0}$ " | 4,6,7, 8,24, 25,26, 27 | Open | 3 | 9 | " | 10 | 44 | " | 17,20 |
| Q94 | $\overline{2E1}$ to $\overline{2E0}$ " | 14,15, 18,19, 33,34, 37,38 | Open | 16 | 20 | " | 10 | 44 | " | 17,20 |
| Q95 | $\overline{1D0}$ to $\overline{1G5}$ " | 4,6,7, 8,24, 25,27 | 3 | 26 | 28 | " | 8 | 31 | " | 17,21 |
| Q96 | $\overline{1D4}$ to $\overline{1G5}$ " | 4,6,7, 24,25, 26,27 | " | 8 | 28 | " | " | " | " | " |
| Q97 | $\overline{1D5}$ to $\overline{1G5}$ " | 4,6,7, 8,24, 26,27 | " | 25 | " | " | " | " | " | " |
| Q98 | $\overline{1D6}$ to $\overline{1G5}$ " | 6,7,8, 24,25, 26,27 | " | 4 | " | " | " | " | " | " |
| Q99 | $\overline{1D7}$ to $\overline{1G5}$ " | 4,6,7, 8,25, 26,27 | " | 24 | " | " | " | " | " | " |
| Q100 | $\overline{2D0}$ to $\overline{2G5}$ " | 14,15, 18,33, 34,37, 38 | 16 | 19 | 40 | " | " | " | " | " |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| Q101 | $\overline{2D4}$ to $\overline{2GS}$ Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | 14,15, 18,19, 34,37, 38 | 16 | 33 | 40 | Open | 8 | 31 | ns | 17,21 |
| Q102 | $\overline{2D5}$ to $\overline{2GS}$ " | 15,18, 19,33, 34,37, 38 | " | 14 | " | " | " | " | " | " |
| Q103 | $\overline{2D6}$ to $\overline{2GS}$ " | 14,15, 18,19, 33,37, 38 | " | 34 | " | " | " | " | " | " |
| Q104 | $\overline{2D7}$ to $\overline{2GS}$ " | 14,18, 19,33, 34,37, 38 | " | 15 | " | " | " | " | " | " |
| Q105 | $\overline{1E1}$ to $\overline{1GS}$ " | 4,6,7, 18,25, 26,27 | 24 | 3 | 28 | " | 7 | 32 | " | 17,20 |
| Q106 | $\overline{2E1}$ to $\overline{2GS}$ " | 14,18, 19,33, 34,37, 38 | 15 | 16 | 40 | " | 7 | 32 | " | " |
| Q107 | $\overline{1D1}$ to $\overline{1A0}$ " | 4,7,8, 24,25, 27 | 3,26 | 6 | 5 | " | 10 | 35 | " | " |
| Q108 | $\overline{1D2}$ to $\overline{1A0}$ " | 4,7,8, 24,25, 26 | 3,6 | 27 | " | " | " | " | " | 17,19 |
| Q109 | $\overline{1D3}$ to $\overline{1A0}$ " | 4,6,8, 24,25, 26 | 3,27 | 7 | " | " | " | " | " | 17,20 |
| Q110 | $\overline{1D4}$ to $\overline{1A0}$ " | 4,6,24, 25,26, 27 | 3,7 | 8 | " | " | " | " | " | 17,20 |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| Q111 | $\overline{ID5}$ to $\overline{IA0}$ Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | 4,6,7, 24,26, 27 | 3,8 | 25 | 5 | Open | 10 | 35 | ns | 17,20 |
| Q112 | $\overline{ID6}$ to $\overline{IA0}$ " | 6,7,8, 24,26, 27 | 3,25 | 4 | " | " | " | " | " | 17,19 |
| Q113 | $\overline{ID7}$ to $\overline{IA0}$ " | 4,6,7, 8,25, 26,27 | 3 | 24 | " | " | " | " | " | 17,20 |
| Q114 | $\overline{ID2}$ to $\overline{IA1}$ " | 4,7,8, 24,25, 26 | 3,6 | 27 | 2 | " | " | " | " | 17,20 |
| Q115 | $\overline{ID3}$ to $\overline{IA1}$ " | 4,8,24, 25,26, 27 | 3,6 | 7 | " | " | " | " | " | 17,20 |
| Q116 | $\overline{ID4}$ to $\overline{IA1}$ " | 4,6,24, 25,26, 27 | 3,7 | 8 | " | " | " | " | " | 17,19 |
| Q117 | $\overline{ID5}$ to $\overline{IA1}$ " | 4,6,8, 24,26, 27 | 3,7 | 25 | " | " | " | " | " | 17,19 |
| Q118 | $\overline{ID6}$ to $\overline{IA1}$ " | 6,7,8, 24,25, 26,27 | 3 | 4 | " | " | " | " | " | 17,20 |
| Q119 | $\overline{ID7}$ to $\overline{IA1}$ " | 4,6,7, 8,25, 26,27 | 3 | 24 | " | " | " | " | " | " |
| Q120 | $\overline{ID4}$ to $\overline{IA2}$ " | 4,6,24, 25,26, 27 | 3,7 | 8 | 23 | " | " | " | " | " |
| Q121 | $\overline{ID5}$ to $\overline{IA2}$ " | 4,6,8, 24,26, 27 | 3,7 | 25 | 23 | " | " | " | " | " |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| Q122 | $\overline{106}$ to $\overline{7A2}$ Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | 6,8,24, 25,26, 27 | 3,7 | 4 | 23 | Open | 10 | 35 | ns | 17,20 |
| Q123 | $\overline{107}$ to $\overline{7A2}$ " | 4,6,7, 8,25, 26,27 | 3 | 24 | " | " | " | " | " | " |
| Q124 | $\overline{201}$ to $\overline{2A0}$ " | 14,15, 18,33, 34,37 | 16,19 | 38 | 39 | " | " | " | " | " |
| Q125 | $\overline{202}$ to $\overline{2A0}$ " | 14,15, 19,33, 34,37 | 16,38 | 18 | " | " | " | " | " | 17,19 |
| Q126 | $\overline{203}$ to $\overline{2A0}$ " | 14,15, 19,33, 34,38 | 16,18 | 37 | " | " | " | " | " | 17,20 |
| Q127 | $\overline{204}$ to $\overline{2A0}$ " | 14,15, 18,19, 34,38 | 16,37 | 33 | " | " | " | " | " | 17,19 |
| Q128 | $\overline{205}$ to $\overline{2A0}$ " | 15,18, 19,34, 37,38 | 16,33 | 14 | " | " | " | " | " | 17,20 |
| Q129 | $\overline{206}$ to $\overline{2A0}$ " | 15,18, 19,33, 37,38 | 14,16 | 34 | " | " | " | " | " | 17,19 |
| Q130 | $\overline{207}$ to $\overline{2A0}$ " | 14,18, 19,33, 34,37, 38 | 16 | 15 | " | " | " | " | " | 17,20 |
| Q131 | $\overline{202}$ to $\overline{2A1}$ " | 14,15, 19,33, 34,37 | 16,18 | 18 | 36 | " | " | " | " | 17,20 |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| Q132 | $\overline{2D3}$ to $\overline{2A1}$ Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | 14,15, 18,19, 33,34 | 16,38 | 37 | 36 | Open | 10 | 35 | ns | 17,20 |
| Q133 | $\overline{2D4}$ to $\overline{2A1}$ " | 14,15, 18,19, 34,38 | 16,37 | 33 | " | " | " | " | " | 17,19 |
| Q134 | $\overline{2D5}$ to $\overline{2A1}$ " | 15,18, 19,33, 34,38 | 16,37 | 14 | " | " | " | " | " | 17,19 |
| Q135 | $\overline{2D6}$ to $\overline{2A1}$ " | 14,15, 18,19, 33,37, 38 | 16 | 34 | " | " | " | " | " | 17,20 |
| Q136 | $\overline{2D7}$ to $\overline{2A1}$ " | 14,18, 19,33, 34,37, 38 | 16 | 15 | " | " | " | " | " | " |
| Q137 | $\overline{2D4}$ to $\overline{2A2}$ " | 14,15, 18,19, 34,38 | 16,37 | 33 | 35 | " | " | " | " | " |
| Q138 | $\overline{2D5}$ to $\overline{2A2}$ " | 15,18, 19,33, 34,38 | 16,37 | 14 | " | " | " | " | " | " |
| Q139 | $\overline{2D6}$ to $\overline{2A2}$ " | 14,15, 18,19, 33,38 | 16,37 | 34 | " | " | " | " | " | " |
| Q140 | $\overline{2D7}$ to $\overline{2A2}$ " | 14,18, 19,33, 34,37, 38 | 16 | 15 | " | " | " | " | " | " |
| Q141 | $\overline{1E1}$ to $\overline{1A0}$ " | 4,6,7, 8,25, 26,27 | 24 | 3 | 5 | " | 7 | 32 | " | " |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| Q142 | $\overline{1E1}$ to $\overline{1A1}$ Propagation delay time ($t_{PHL}$), $C_L$ = 150 pf | 4,6,7, 8,25, 26,27 | 24 | 3 | 2 | Open | 7 | 32 | ns | 17,20 |
| Q143 | $\overline{1E1}$ to $\overline{1A2}$ " | 4,6,7, 8,25, 26,27 | 24 | 3 | 23 | " | " | " | " | " |
| Q144 | $\overline{2E1}$ to $\overline{2A0}$ " | 14,18, 19,33, 34,37, 38 | 15 | 16 | 39 | " | " | " | " | " |
| Q145 | $\overline{2E1}$ to $\overline{2A1}$ " | 14,18, 19,33, 34,37, 38 | 15 | 16 | 36 | " | " | " | " | " |
| Q146 | $\overline{2E1}$ to $\overline{2A2}$ " | 14,18, 19,33, 34,37, 38 | 15 | 16 | 35 | " | " | " | " | " |
| Q147 | $\overline{1D1}$ to 1A0 " | 4,7,8, 24,25, 27 | 3,26 | 6 | 30 | 5 | 13 | 68 | " | 17,19 |
| Q148 | $\overline{1D7}$ to 1A0 " | 4,6,7, 8,25, 26,27 | 3 | 24 | 30 | 5 | " | " | " | " |
| Q149 | $\overline{1D2}$ to 1A1 " | 4,7,8, 24,25, 26 | 3,6 | 27 | 31 | 2 | " | " | " | " |
| Q150 | $\overline{1D6}$ to 1A1 " | 6,7,8, 24,25, 26,27 | 3 | 4 | 31 | 2 | " | " | " | " |
| Q151 | $\overline{1D4}$ to 1A2 " | 4,6,24, 25,26, 27 | 3,7 | 8 | 11 | 23 | " | " | " | " |





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TABLE XIV. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| Q152 | $\overline{1D5}$ to 1A2 Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | 4,6,8, 24,26, 27 | 3,7 | 25 | 11 | 23 | 13 | 68 | ns | 17,19 |
| Q153 | $\overline{2D1}$ to 2A0 " | 14,15, 18,33, 34,37 | 16,19 | 38 | 32 | 39 | " | " | " | " |
| Q154 | $\overline{2D7}$ to 2A0 " | 14,18, 19,33, 34,37, 38 | 16 | 15 | 32 | 39 | " | " | " | " |
| Q155 | $\overline{2D2}$ to 2A1 " | 14,15, 19,33, 34,37 | 16,38 | 18 | 12 | 36 | " | " | " | " |
| Q156 | $\overline{2D6}$ to 2A1 " | 14,15, 18,19, 33,37, 38 | 16 | 34 | 12 | 36 | " | " | " | " |
| Q157 | $\overline{2D4}$ to 2A2 " | 14,15, 18,19, 34,38 | 16,37 | 33 | 13 | 35 | " | " | " | " |
| Q158 | $\overline{2D5}$ to 2A2 " | 15,18, 19,33, 34,38 | 16,37 | 14 | 13 | 35 | " | " | " | " |
| Q159 | $\overline{1E1}$ to 1A0 " | 4,6,7, 8,25, 26,27 | 24 | 3 | 30 | 5 | 10 | 56 | " | " |
| Q160 | $\overline{1E1}$ to 1A1 " | 4,6,7, 8,25, 26,27 | 24 | 3 | 31 | 2 | " | " | " | " |
| Q161 | $\overline{1E1}$ to 1A2 " | 4,6,7, 8,25, 26,27 | 24 | 3 | 11 | 23 | " | " | " | " |



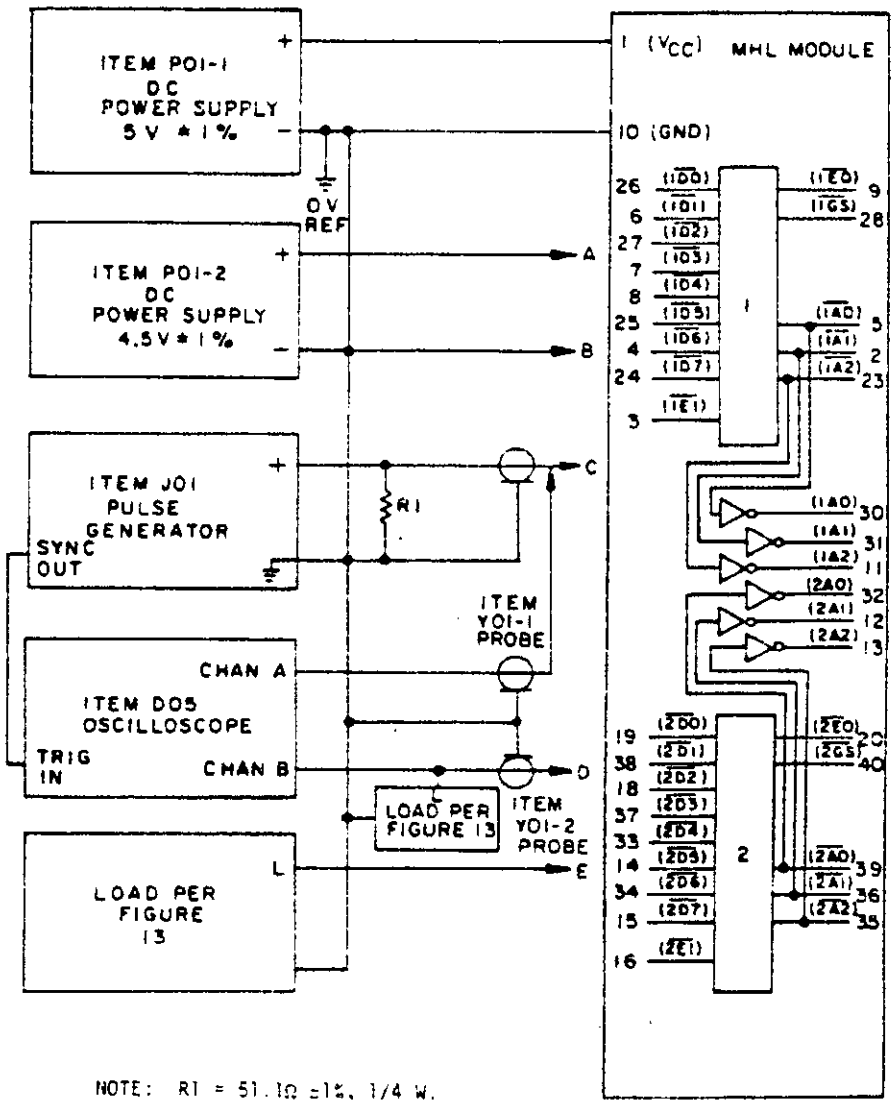


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TABLE XIV. Transition time ($t_{THL}$ and $t_{TLH}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test continued

| Test condition | Parameter | Connect | | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| Q162 | $\overline{2E1}$ to 2A0 Propagation delay time ($t_{PHL}$), $t_L$ = 150 pF | 14,18, 19,33, 34,37 38 | 15 | 16 | 12 | 39 | 10 | 56 | ns | 17,19 |
| Q163 | $\overline{2E1}$ to 2A1 | 14,18, 19,33, 34,37, 38 | " | " | 12 | 46 | " | " | " | " |
| Q164 | $\overline{2E1}$ to 2A2 | 14,18, 19,33, 34,37, 38 | " | " | 13 | 45 | " | " | " | " |

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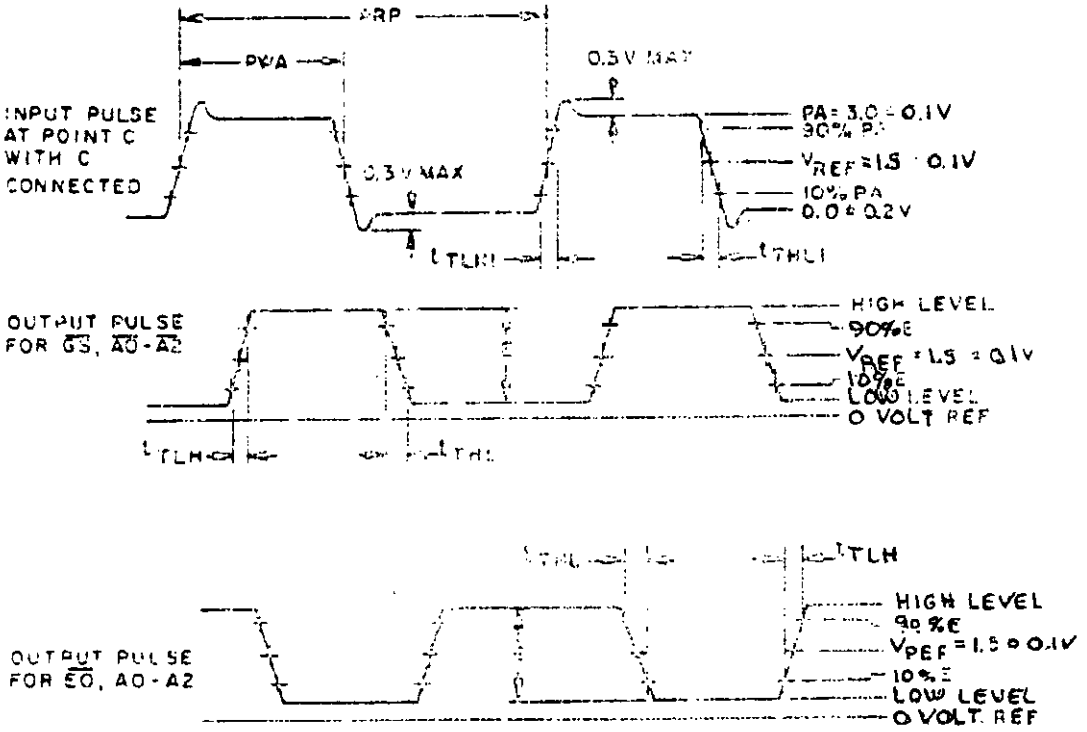



NOTE: R1 = 51.1Ω ±1%, 1/4 W.

FIGURE 17. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test circuit for table XIV.



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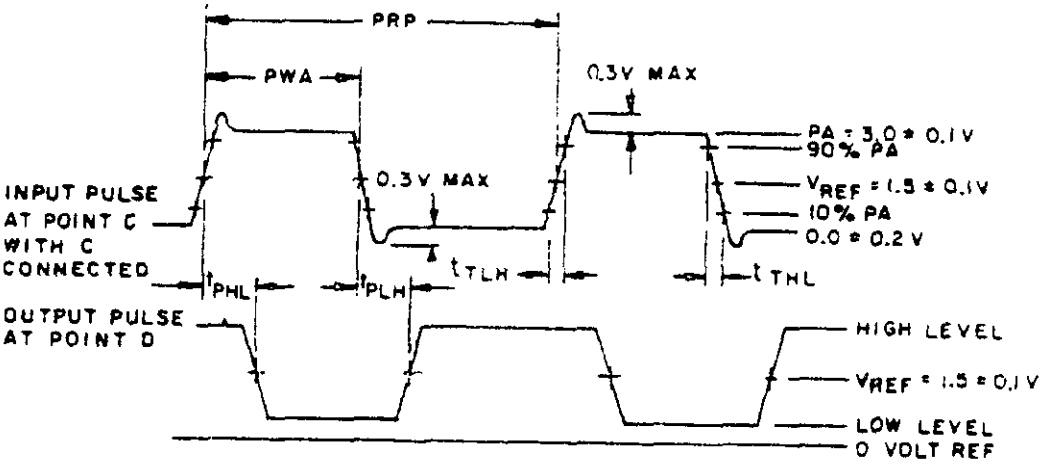



NOTES:
1. Input pulse parameters:
   $t_{TLH}$ = $t_{THL}$ = 6 ±1 ns
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements specified on table XIV, verify that the output pulse high level is ≥ 2.4 V and the low level is ≤ 0.4 V.
3. The lead length of electrical conductors between resistor R1 and input pins, between probes V01-1 and V01-2 and output pins; one between load circuits and output pins, shall be less than 2 inches (50.8 mm).
4. All voltages are referenced to ground.

FIGURE 18. Transition time ($t_{TLH}$ and $t_{THL}$) test circuit waveforms for table XIV.



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Check the source to verify that this is the current version before use.

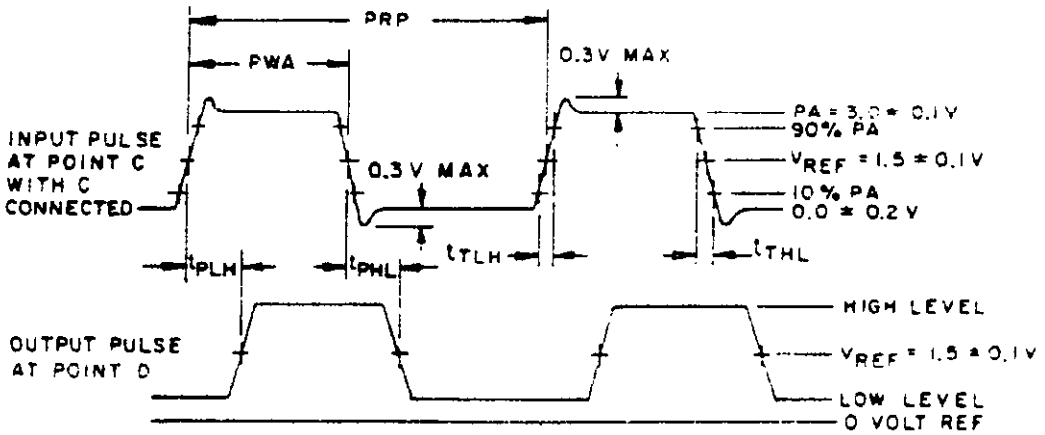



NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 6 \pm 1$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements specified in table XIV, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.
3. The lead length of electrical conductors between resistor R1 and input pins; between probes YO1-1 and YO1-2 and output pins; and between load circuits and output pins, shall be less than 2 inches (50.8 mm).
4. All voltages are referenced to ground.

FIGURE 19. Propagation delay time ($t_{PLP}$ and $t_{PHL}$) test circuit waveforms for table XIV.



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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 6 \pm 1$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements specified in table XIV, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.
3. The lead length of electrical conductors between resistor R1 and input pins; between probes YO1-1 and YO1-2 and output pins; and between load circuits and output pins, shall be less than 2 inches (50.8 mm).
4. All voltages are referenced to ground.

FIGURE 20. Propagation delay time ($t_{PLH}$ and $t_{PHL}$) test circuit waveforms for table XIV.



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TABLE XV. Output crosstalk (low-level, high-level, and duration) test.

| Test condition | Parameter | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| R1 | $\overline{TGS}$ Output crosstalk, low level | 4,6,7,8,14, 18,19,25, 27,33,34, 37,38 | 3,16,26 | 15,24 | 28 | 0 | 800 | mV | 21,22 |
| R2 | $\overline{TA0}$ " | 4,7,8,14, 18,19,25, 26,27,33, 34,37,38 | 3,6,16 | " | 5 | " | 650 | " | " |
| R3 | $\overline{TA1}$ " | 4,6,7,8,14, 18,19,25, 26,33,34, 37,38 | 3,16,27 | " | 2 | " | " | " | " |
| R4 | $\overline{TA2}$ " | 4,6,7,14, 18,19,25, 26,27,33, 34,37,38 | 3,8,16 | " | 23 | " | " | " | " |
| R5 | 1A0 " | 6,7,8,14, 18,19,24, 25,26,27, 33,34,37, 38 | 3,16 | 4,15 | 30 | " | " | " | " |
| R6 | 1A1 " | 4,6,7,8,14, 18,19,24, 26,27,33, 34,37,38 | 3,16 | 15,25 | 31 | " | " | " | " |
| R7 | 1A2 " | 4,6,8,14, 18,19,24, 25,26,27, 33,34,37, 38 | 3,16 | 7,15 | 11 | " | " | " | " |
| R8 | $\overline{1E0}$ " | 4,6,7,8,14, 18,19,24, 25,26,27, 33,34,37, 38 | 3,16 | 15 | 9 | " | " | " | " |





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TABLE XV. Output crosstalk (low-level, high-level, and duration) test - Continued.

| Test condition | Parameter | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| R9 | $\overline{2GS}$ Output crosstalk, low level | 4,6,7,8,14, 18,25,26, 27,33,34, 37,38 | 3,16,19 | 15,24 | 40 | 0 | 800 | mV | 21,22 |
| R10 | $\overline{2A0}$ " | 4,6,7,8,14, 18,19,25, 26,27,33, 34,37 | 3,16,38 | " | 39 | " | 650 | " | " |
| R11 | $\overline{2A1}$ " | 4,6,7,8,14, 19,25,26, 27,33,34, 37,38 | 3,16,18 | " | 36 | " | " | " | " |
| R12 | $\overline{2A2}$ " | 4,6,7,8,14, 18,19,25, 26,27,34, 37,38 | 3,16,33 | " | 35 | " | " | " | " |
| R13 | 2A0 " | 4,6,7,8,14, 15,18,19, 25,26,27, 33,37,38 | 3,16 | 24,34 | 32 | " | " | " | " |
| R14 | 2A1 " | 4,6,7,8,15, 18,19,25, 26,27,33, 34,37,38 | " | 14,24 | 12 | " | " | " | " |
| R15 | 2A2 " | 4,6,7,8,14, 15,18,19, 25,26,27, 33,34,38 | " | 24,37 | 13 | " | " | " | " |
| R16 | $\overline{2F0}$ " | 4,6,7,8,14, 15,18,19, 25,26,27, 33,34,37, 38 | " | 24 | 20 | " | " | " | " |





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TABLE XV. Output crosstalk (low-level, high-level, and duration) test - Continued.

| Test condition | Parameter | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| R17 | $\overline{TCS}$ Output crosstalk, duration | 4,5,7,8,14, 13,19,25, 27,33,34, 37,38 | 3,16,26 | 15,24 | 28 | 0 | 400 | mV | 21,23 |
| R18 | $\overline{TA0}$ " | 4,7,8,14, 18,19,25, 26,27,33, 34,37,38 | 3,6,16 | " | 5 | " | " | " | " |
| R19 | $\overline{TA1}$ " | 4,5,7,8,14, 18,19,25, 26,33,34, 37,38 | 3,16,27 | " | 2 | " | " | " | " |
| R20 | $\overline{TA2}$ " | 4,6,7,14, 18,19,25, 26,27,33, 34,37,38 | 3,8,16 | " | 23 | " | " | " | " |
| R21 | TA0 " | 6,7,8,14, 18,19,24, 25,26,27, 33,34,37, 38 | 3,16 | 4,15 | 30 | " | " | " | " |
| R22 | TA1 " | 4,6,7,8,14, 18,19,26, 27,33,34, 37,38,24 | " | 15,25 | 31 | " | " | " | " |
| R23 | TA2 " | 4,6,8,14, 18,19,24, 25,26,27, 31,34,37, 38 | " | 7,15 | 11 | " | " | " | " |





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TABLE XV. Output crosstalk (low-level, high-level, and duration) test - Continued.

| Test condition | Parameter | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| R24 | $\overline{TEO}$ Output crosstalk, duration | 4,6,7,8,14, 18,19,24, 25,26,27, 33,34,37, 38 | 3,16 | 15 | 9 | 0 | 400 | mV | 21,23 |
| R25 | $\overline{2GS}$ " | 4,6,7,8,14, 18,25,26, 27,33,34, 37,38 | 3,16,19 | 15,24 | 40 | " | " | " | " |
| R26 | 2A0 " | 4,6,7,8,14, 18,19,25, 26,27,33, 34,37 | 3,16,38 | " | 39 | " | " | " | " |
| R27 | $\overline{2A1}$ " | 4,6,7,8,14, 19,25,26, 27,33,34, 37,38 | 3,16,18 | " | 36 | " | " | " | " |
| R28 | $\overline{2A2}$ " | 4,6,7,8,14, 18,19,25, 26,34,37, 38,27 | 3,16,33 | " | 35 | " | " | " | " |
| R29 | 2A0 " | 4,6,7,8,14, 15,18,19, 25,26,27, 33,37,38 | 3,16 | 24,34 | 38 | " | " | " | " |
| R30 | 2A1 " | 4,6,7,8,15, 18,19,25, 26,27,33, 34,37,38 | " | 14,24 | 12 | " | " | " | " |
| R31 | 2A2 " | 4,6,7,8,14, 15,18,19, 25,26,27, 33,34,38 | " | 24,37 | 11 | " | " | " | " |





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TABLE XV. Output crosstalk (low-level, high-level, and duration) test - Continued.

| Test condition | Parameter | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| R32 | $\overline{ZEO}$ Output crosstalk, duration | 4,6,7,8,14, 15,18,19, 25,26,27, 33,34,37, 38 | 3,16 | 24 | 20 | 0 | 400 | mV | 21,23 |
| S1 | $\overline{TGS}$ Output crosstalk, high level | 3,4,6,7,8, 14,18,19, 24,25,26, 27,33,34, 37,38 | 16 | 15 | 28 | " | " | " | 21,22 |
| S2 | $\overline{TAO}$ " | 6,7,8,14, 18,19,24, 25,26,27, 33,34,37, 38 | 3,16 | 4,15 | 5 | " | " | " | " |
| S3 | $\overline{TAT}$ " | 4,6,7,8,14, 18,19,24, 26,27,33, 34,37,38 | 3,16 | 15,25 | 2 | " | " | " | " |
| S4 | $\overline{TA2}$ " | 4,6,8,14, 18,19,24, 25,26,27, 33,34,37, 38 | 3,16 | 7,15 | 23 | " | " | " | " |





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TABLE XV. Output crosstalk (low-level, high-level, and duration) test - Continued.

| Test condition | Parameter | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| S5 | 1A0 Output crosstalk, high level | 4,7,8,14, 18,19,25, 26,27,33, 34,37,38 | 3,5,16 | 15,24 | 30 | 0 | 400 | mV | 21,22 |
| S6 | 1A1 " | 4,6,7,8,14, 18,19,25, 26,33,34, 37,38 | 3,16,27 | 15,24 | 31 | " | " | " | " |
| S7 | 1A2 " | 4,6,7,14, 18,19,25, 26,27,33, 34,37,38 | 3,8,16 | 15,24 | 11 | " | " | " | " |
| S8 | $\overline{1E0}$ " | 4,6,7,8,14, 18,19,25, 26,27,33, 34,37,38 | 16,24 | 3,15 | 9 | " | " | " | " |
| S9 | 2GS " | 4,6,7,8,14, 15,16,18, 19,25,26, 27,33,34, 37,38 | 3 | 24 | 40 | " | " | " | " |
| S10 | $\overline{2A0}$ " | 4,6,7,8,14, 15,18,19, 25,26,27, 33,37,38 | 3,16 | 24,34 | 39 | " | " | " | " |
| S11 | $\overline{2A1}$ " | 4,6,7,8,15, 18,19,25, 26,27,33, 34,37,38 | 3,16 | 14,24 | 36 | " | " | " | " |
| S12 | $\overline{2A2}$ " | 4,6,7,8,14, 15,18,19, 25,26,27, 33,34,38 | 3,16 | 24,37 | 35 | " | " | " | " |





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TABLE XV. Output crosstalk (low-level, high-level, and duration) test - Continued.

| Test condition | Parameter | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| S13 | 2A0 Output crosstalk, high level | 4,6,7,8,14, 18,19,25, 26,27,33, 34,37 | 3,16,38 | 15,24 | 32 | 0 | 400 | mV | 21,22 |
| S14 | 2A1 " | 4,5,7,8,14, 19,25,26, 27,33,34, 37,38 | 3,16,18 | 15,24 | 12 | " | " | " | " |
| S15 | 2A2 " | 1,5,7,8,14, 18,19,25, 26,27,34, 37,38 | 3,16,33 | 15,24 | 13 | " | " | " | " |
| S16 | $\overline{2E0}$ " | 4,6,7,8,14, 18,19,25, 26,27,33, 34,37,38 | 3,15 | 16,24 | 20 | " | " | " | " |

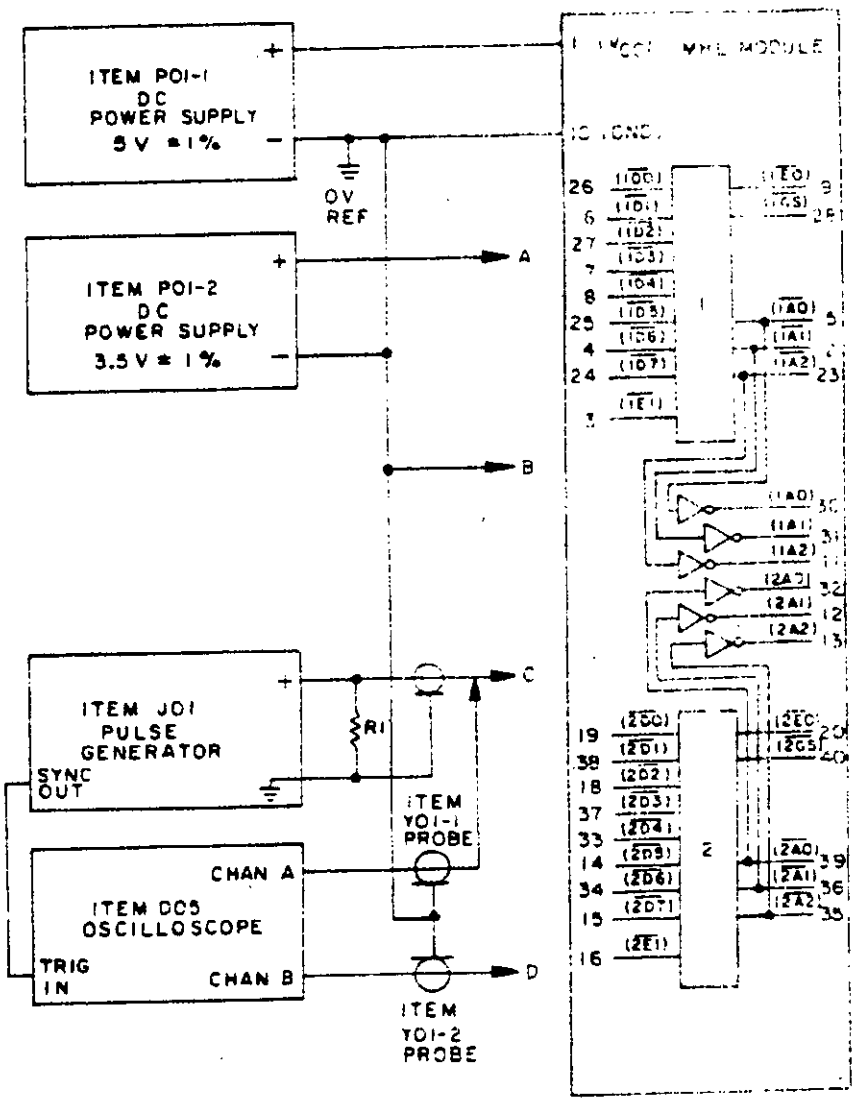
TABLE XVI. Power supply transient amplitude test.

| Test condition | Parameter | Initial limits 25°C±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|
| | | Min | Max | | |
| T | Power supply transient amplitude | 0 | 400 | mV | 24,25 |





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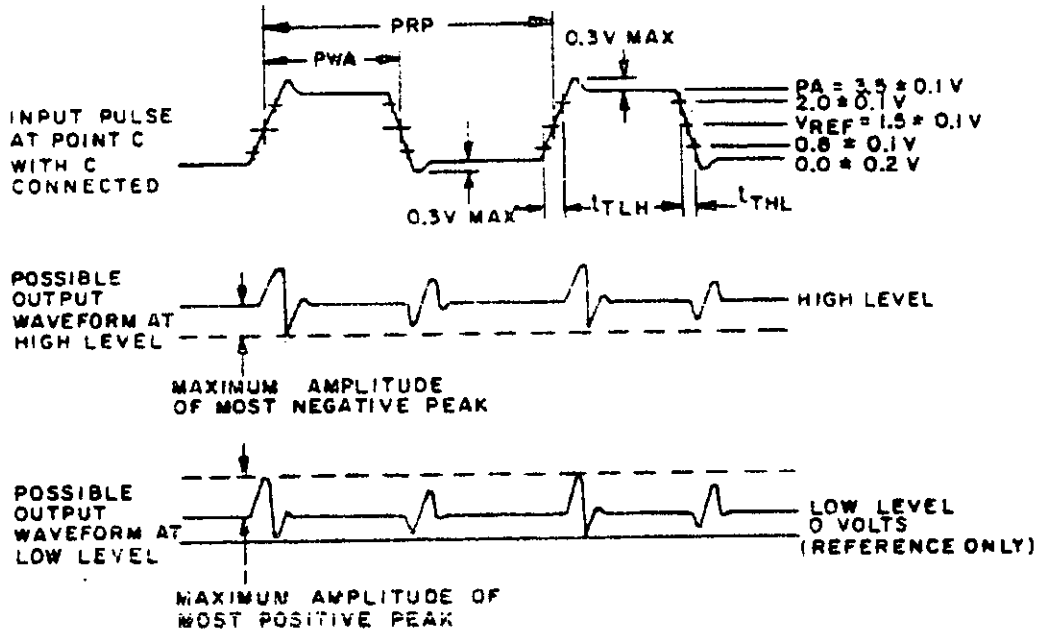



NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. Load all outputs with load as specified on figure 13 with the following exception: Capacitor $C_L$ usec in MHL load circuit is 50 pF for test table XV.

FIGURE 21. Output crosstalk (low-level, high-level, and duration test circuit for table XV.



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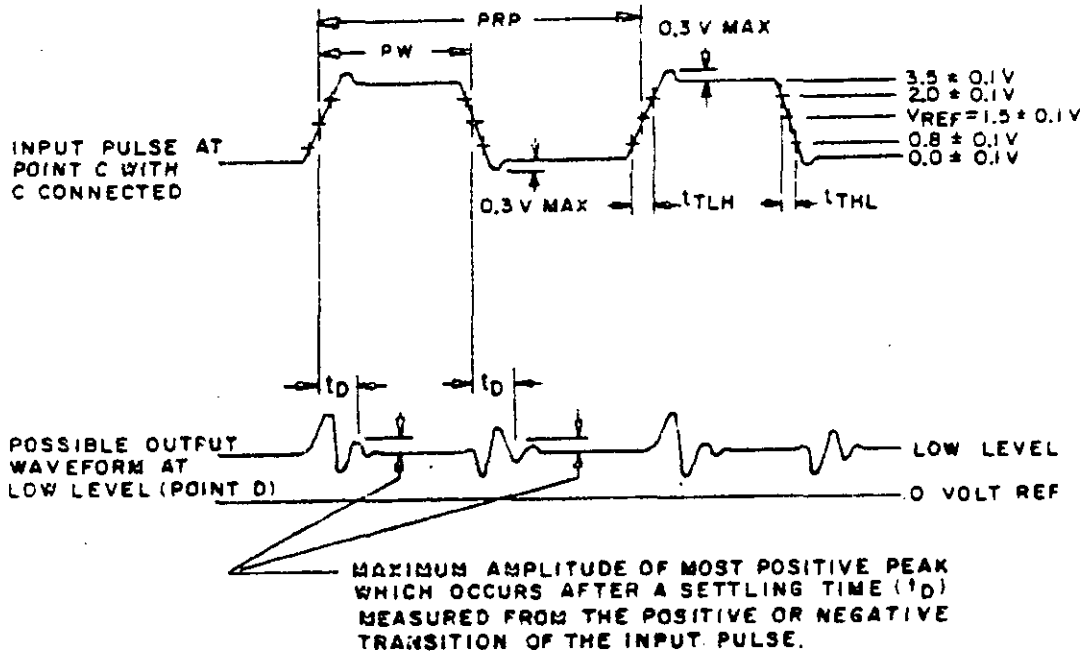



NOTES:
1. Input pulse parameters:
   $t_{THL} = t_{TLH} = 3.5 \pm 1.5$ ns (verified at each input pin).
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. The lead length of electrical conductors between resistor R1 and input pins; between probes YOI-1 and YOI-2 and input and output pins; between load circuits and output pins; and between ground leads of probes YOI-1 and YOI-2 and test fixture and module ground pin, shall be less than 2 inches (50.8 mm).

FIGURE 22. Output crosstalk (low-level and high-level) test circuit waveforms for table XV.



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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin).
   PW = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
   $t_D$ = 76 ns.
2. The lead length of electrical conductors between resistor R1 and input pins, between probes Y01-1 and Y01-2 and input and output pins; between load circuits and output pins; and between ground leads of probes Y01-1 and Y01-2 and test fixture and module ground pins, shall be less than 2 inches (50.8 mm).

FIGURE 23. Output crosstalk (duration) test circuit waveforms for table XV.



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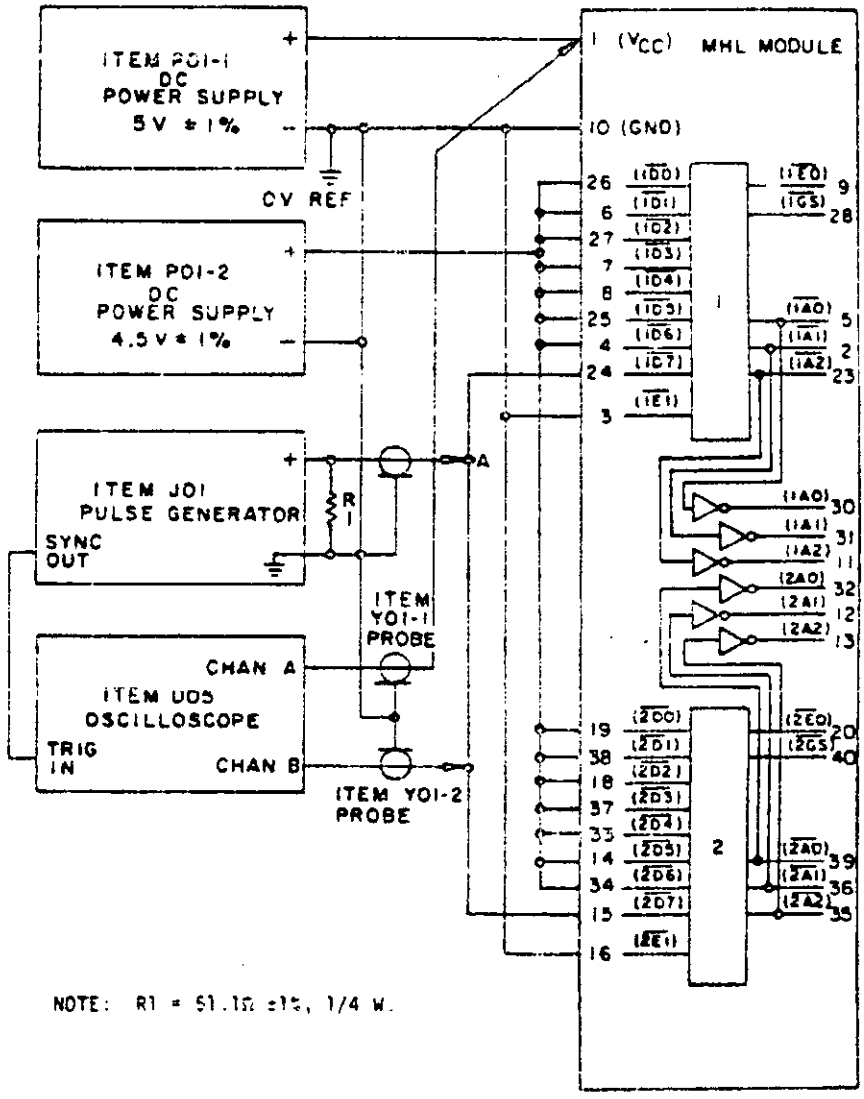
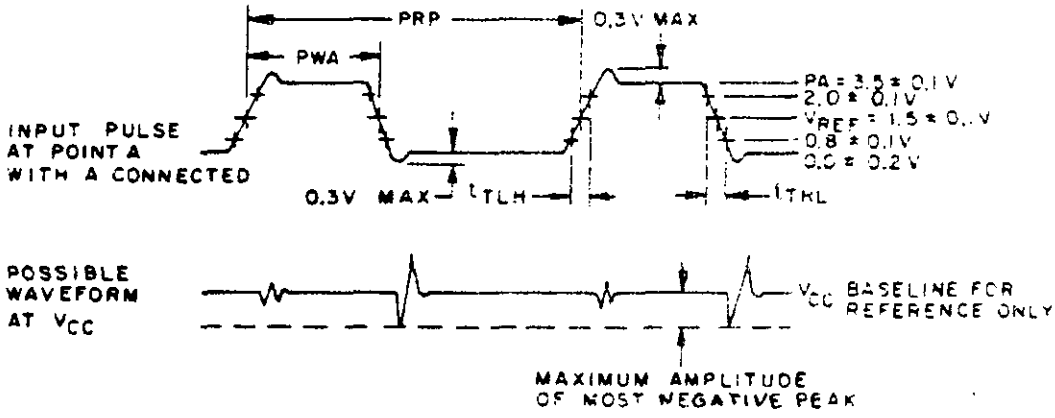



FIGURE 24. Power supply transient amplitude test circuit for table XVI.



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NOTE: Input pulse parameters:
$t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin
PWA = 500 ±50 ns.
PRP = 1,000 ±100 ns.

FIGURE 25. Power supply transient amplitude test circuit waveform for table XVI



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TABLE XVII. Power supply filter capacitance test.

| Test condition | Parameter | Connect | | Initial limits 25°C±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| U | Power supply filter capacitance | Open | 1 | 1.6 | 3.0 | μF | 26 |

TABLE XVIII. Input Capacitance test.

| Test condition | Parameter | Connect | | Initial limits 25°C±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| V1 | $\overline{100}$ Input capacitance | 26 | Open | 5 | 15 | pF | 26 |
| V2 | $\overline{101}$ " | 6 | " | " | " | " | " |
| V3 | $\overline{102}$ " | 27 | " | " | " | " | " |
| V4 | $\overline{103}$ " | 7 | " | " | " | " | " |
| V5 | $\overline{104}$ " | 8 | " | " | " | " | " |
| V6 | $\overline{105}$ " | 25 | " | " | " | " | " |
| V7 | $\overline{106}$ " | 4 | " | " | " | " | " |
| V8 | $\overline{107}$ " | 24 | " | " | " | " | " |
| V9 | $\overline{1E1}$ " | 3 | " | " | " | " | " |
| V10 | $\overline{200}$ " | 19 | " | " | " | " | " |
| V11 | $\overline{201}$ " | 38 | " | " | " | " | " |
| V12 | $\overline{202}$ " | 18 | " | " | " | " | " |
| V13 | $\overline{203}$ " | 37 | " | " | " | " | " |
| V14 | $\overline{204}$ " | 33 | " | " | " | " | " |
| V15 | $\overline{205}$ " | 14 | " | " | " | " | " |





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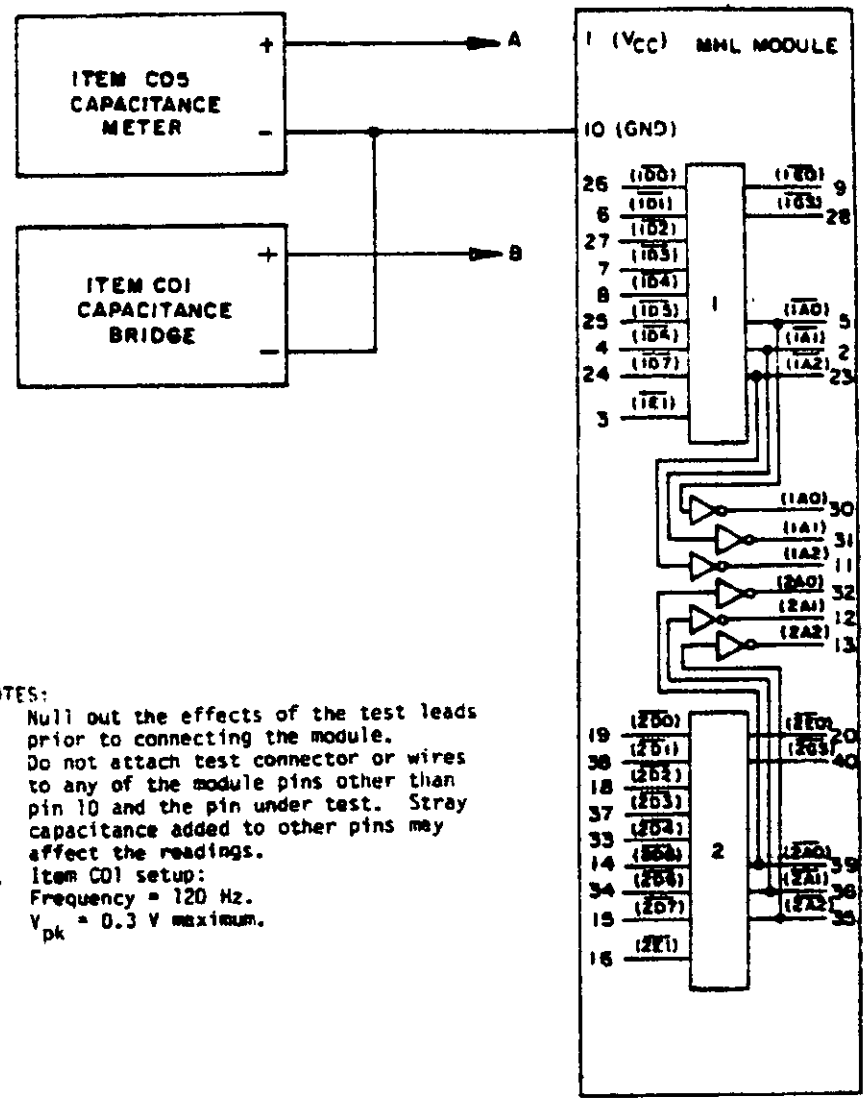
TABLE XVIII. Input Capacitance test.

| Test condition | Parameter | Connect | | Initial limits 25°C±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| V16 | $\overline{206}$ Input capacitance | 34 | Open | 5 | 15 | pF | 26 |
| V17 | $\overline{207}$ " | 15 | " | " | " | " | " |
| V18 | $\overline{2E1}$ " | 16 | " | " | " | " | " |





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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item C01 setup:
   Frequency = 120 Hz.
   $V_{pk}$ = 0.3 V maximum.

FIGURE 26. Power supply filter and input capacitance test circuit for tables XVII and XVIII.



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